

U4: The Network Layer

What does Network Layer do?

It is responsible for:

- Delivering packets from source → destination
- Across multiple networks and routers

Note: It is the lowest layer that handles end-to-end communication

- Data link layer → works node-to-node
- Network layer → works source to destination

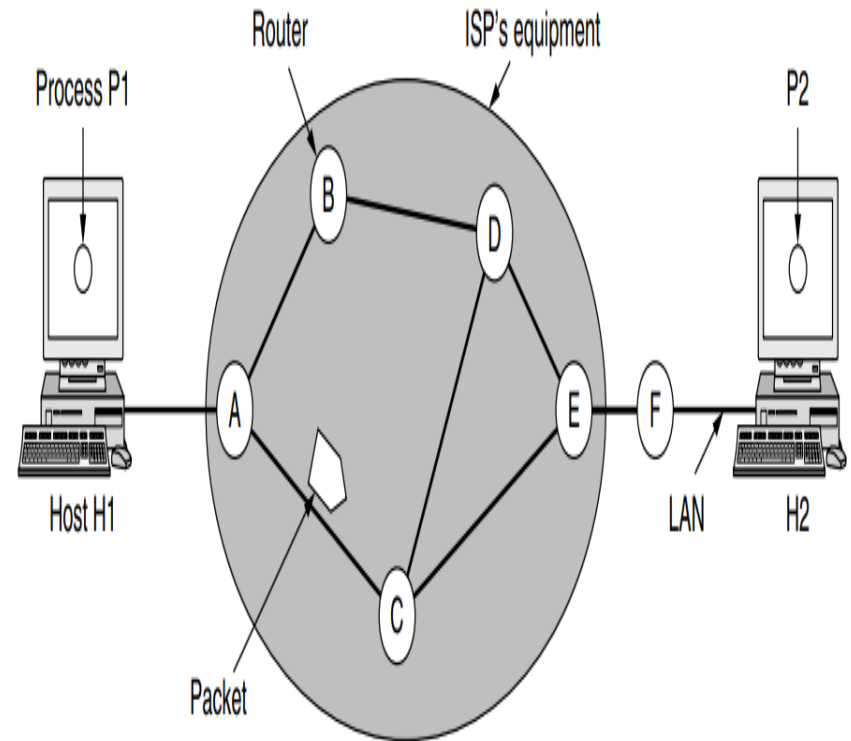
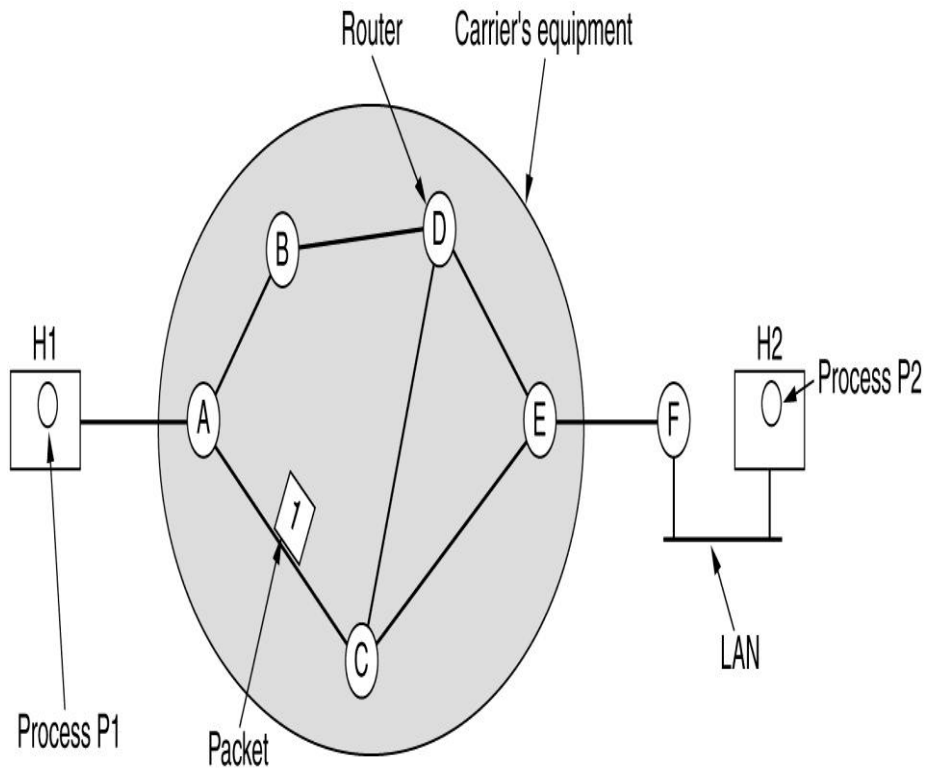
Example:

- Data link = local courier within city
- Network layer = courier across cities/countries

Network Layer Design Issues

- Store-and-Forward Packet Switching
- Services Provided to the Transport Layer
- Implementation of Connectionless Service
- Implementation of Connection-Oriented Service
- Comparison of Virtual-Circuit and Datagram Subnets

Store-and-Forward Packet Switching



The environment of the network layer protocols.

Store-and-Forward Packet Switching

Working:

1. Host sends packet → Router
2. Router:
 1. Stores packet
 2. Checks error (checksum)
3. Forwards to next router
4. Process repeats

Flow: Host → Router A → Router B → Router C → Destination

Ex. Courier system:

- Package stops at multiple hubs before reaching destination

Note: Each router processes packet completely before forwarding

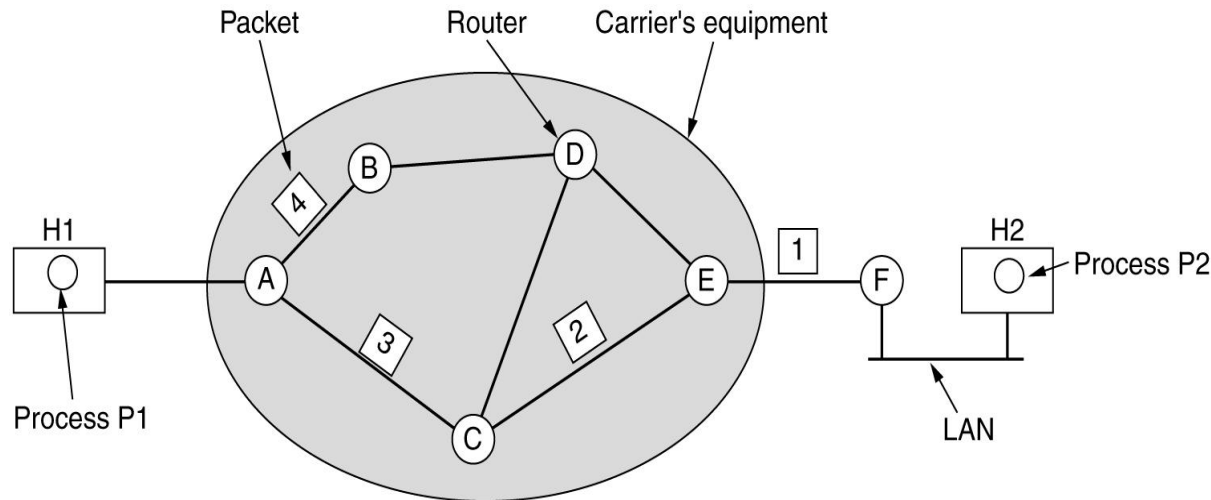
Services Provided to the Transport Layer

1. Services independent of router technology.
2. Transport layer shielded from number, type, topology of routers.
3. Network addresses available to transport layer use uniform numbering plan even across LANs and WANs

What kind of services the network layer provides to the transport layer and its goals:

1. Independent of hardware
2. Hide complexity of network
3. Uniform addressing

Implementation of Connectionless Service (Datagram)



Working:

1. Message → divided into packets
2. Each packet:
 - 2.1 Routed independently
 - 2.2 May take different paths

From diagram:

- Packet 1 → A → C → E → F
- Packet 4 → A → B → D → F

Different routes for same destination

Properties:

- No route setup
- Each packet carries full address
- Routing done at every hop

Note: Packets may arrive:

- Out of order
- With delay variation

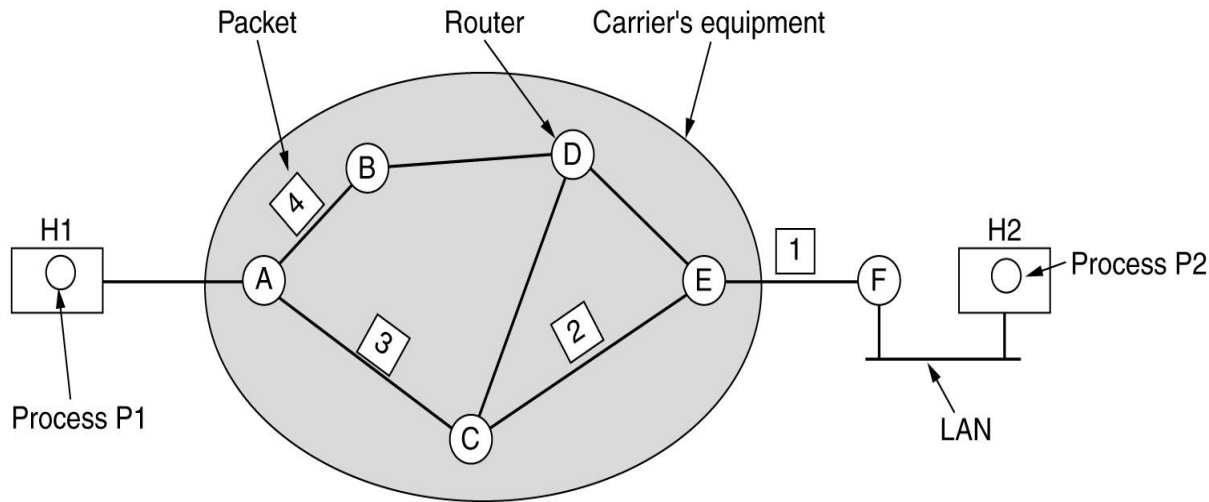
A's table			
initially	later	C's table	E's table
A -	A -	A A	A C
B B	B B	B A	B D
C C	C C	C -	C C
D B	D B	D D	D D
E C	E B	E E	E -
F C	F B	F E	F F
Dest. Line			

Example:

Internet (IP protocol)

Routing within a diagram subnet.

Implementation of Connectionless Service (Datagram)



A's table

initially	later
A -	A -
B B	B B
C C	C C
D B	D B
E C	E B
F C	F B

C's table

A A
B A
C -
D D
E E
F E

E's table

A C
B D
C C
D D
E -
F F

Dest. Line

Concept:

- No setup
- Each packet independent

Features:

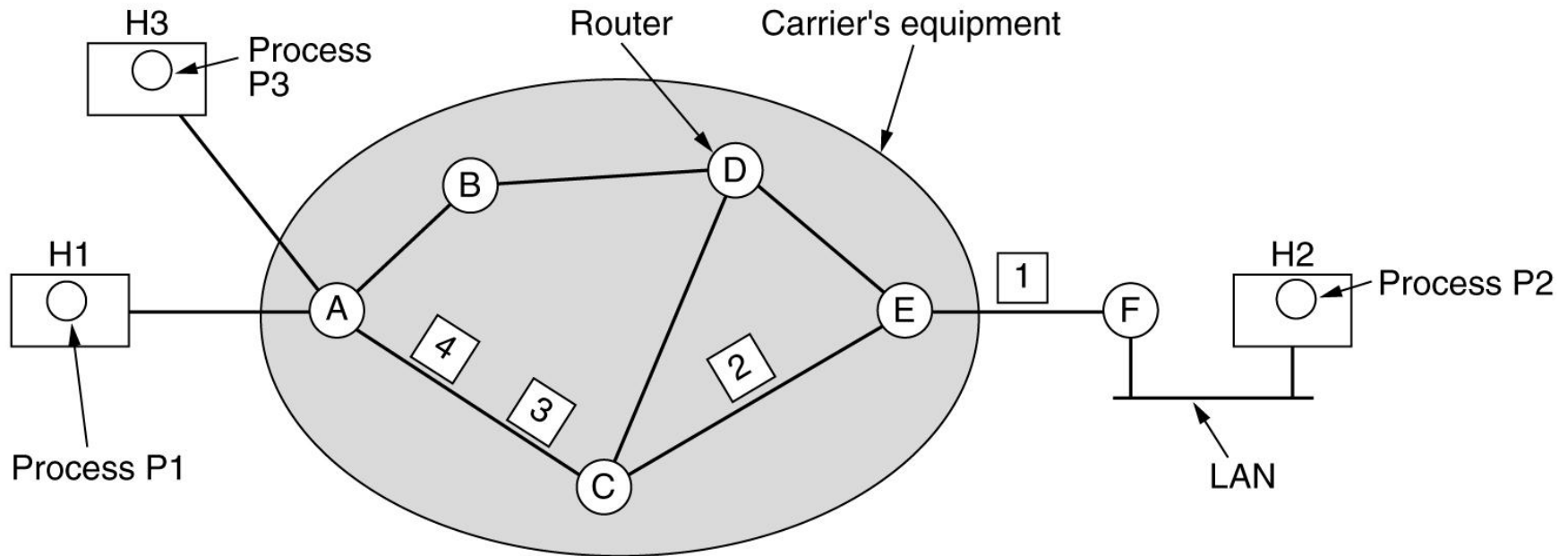
- No guarantee
- No order
- No reliability

Example:

Internet traffic (IP packets)

Routing within a diagram subnet.

Implementation of Connection-Oriented Service



A's table		C's table		E's table	
H1	1	A	1	C	1
H3	1	A	2	C	2
In		Out		Out	

Routing within a virtual-circuit subnet.

Implementation of Connection-Oriented Service

Working:

Step 1: Setup Phase

- Path selected
- Stored in routers

Step 2: Data Transfer

- All packets follow same path

Step 3: Termination

- Connection closed

Key Feature:

Each packet carries:

VC number (not full address)

Understanding Example:

Railway reservation:

- Route fixed before journey

Concept:

- Setup path first
- Then send data

Features:

- Reliable
- Ordered
- QoS possible

Example:

Telephone network, MPLS

Label Switching (MPLS)

- Routers change labels
- Faster forwarding

Note: Label switching. An example of a connection-oriented network service is **MPLS (MultiProtocol Label Switching)**. It is used within ISP networks in the Internet, with IP packets wrapped in an **MPLS header having a 20-bit connection identifier or label**. MPLS is often hidden from customers, with the ISP establishing long-term connections for large amounts of traffic, but it is increasingly being used to help when quality of service is important but also with other ISP traffic management tasks.

Comparison of Virtual-Circuit and Datagram Subnets

Issue	Datagram subnet	Virtual-circuit subnet
Circuit setup	Not needed	Required
Addressing	Each packet contains the full source and destination address	Each packet contains a short VC number
State information	Routers do not hold state information about connections	Each VC requires router table space per connection
Routing	Each packet is routed independently	Route chosen when VC is set up; all packets follow it
Effect of router failures	None, except for packets lost during the crash	All VCs that passed through the failed router are terminated
Quality of service	Difficult	Easy if enough resources can be allocated in advance for each VC
Congestion control	Difficult	Easy if enough resources can be allocated in advance for each VC

Based on this discussion
Which is used in Internet?
Datagram or Virtual Circuit (Why?)

Based on this discussion
Which is used in Internet?

Ans: Datagram

ROUTING vs FORWARDING

Routing:

- Decide path
- Done by algorithm

Forwarding:

- Move packet
- Based on routing table

Example:

- Routing = Google Maps
- Forwarding = Driving

Routing Algorithms

Routing is a process of transferring the packets from source machine to destination machine, while **routing algorithms** (RA) are software responsible for deciding which outgoing line an incoming packet should be transmitted on.

The **purpose of RA** is to decide which route is to be followed by a packet on the basis of following parameters:

- Availability of channels (paths/links/routes)
- Link transmission delay
- Traffic intensity, and
- Capacity (bandwidth) of the link

Routing Algorithms

On the basis of **nature of information** used in the algorithm, Routing Algorithms are categorized into two classes:

- **Non-Adaptive:** These do not base their routing decisions on the measurements/estimates of current traffic or topology, instead the choice of route to get from node 'i' to node 'j' is computed in advance or off-line, and hence also called as 'Static Routing' or 'Pre-determined Routing'.
- **Adaptive:** It attempts to change their routing decisions to reflect changes in topology and the current traffic.

On the basis of **scope of information** used in the algorithm, Routing Algorithms are classified into three types:

- **Centralized Routing:** The global algorithm uses information collected from the entire subnet in an attempt to make optimal decision.
- **Isolated Routing:** The local algorithm runs separately on each IMP and uses information only available there e.g., queue length.
- **Distributed Routing:** These use information available locally as well as information available at their neighbors.

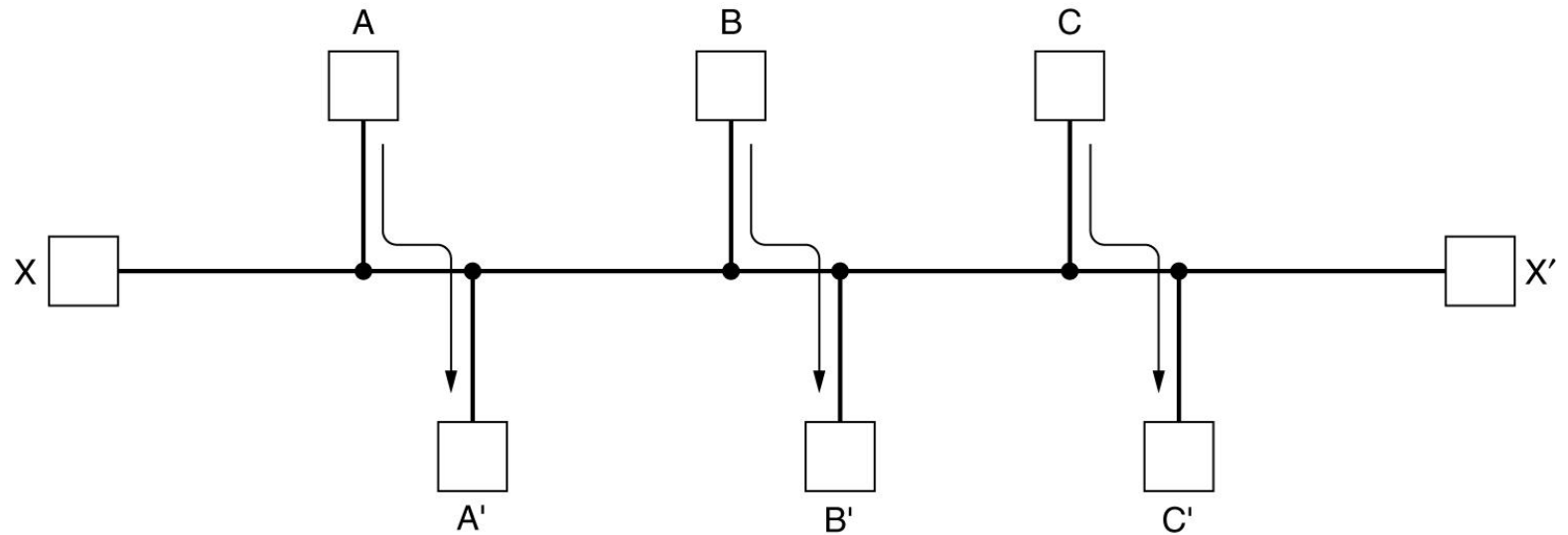
Routing Algorithms

- **The Optimality Principle**
- **Shortest Path Routing (DIJKSTRA)**
- Flooding
- **Distance Vector Routing**
- Link State Routing
- Hierarchical Routing
- Broadcast Routing
- Multicast Routing
- Routing for Mobile Hosts
- Routing in Ad Hoc Networks

Routing Algorithm Properties

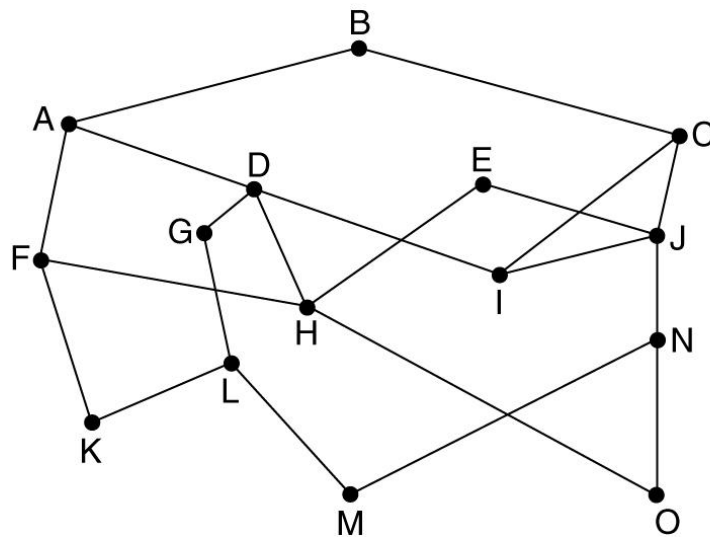
- Correctness
- Simplicity
- Robustness
- Stability
- Fairness
- Efficiency

Routing Algorithms (2)

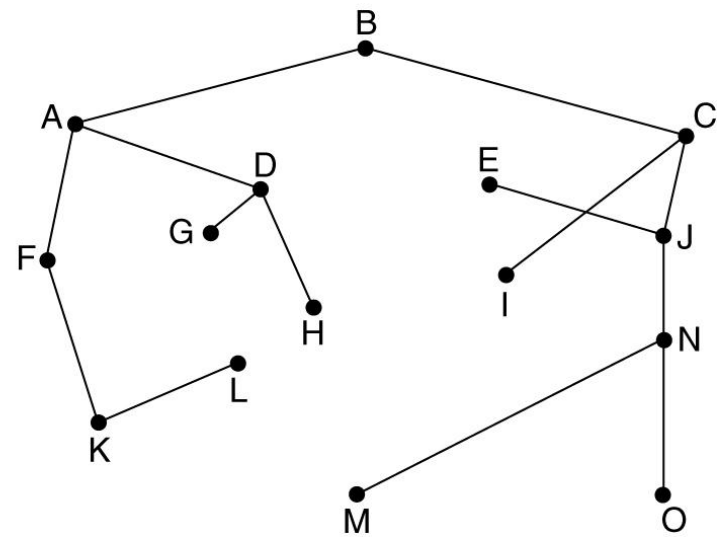


Conflict between fairness and optimality.

The Optimality Principle



(a)



(b)

DAG (Directed Acyclic Graph)

(a) A subnet. (b) A sink tree for router B.

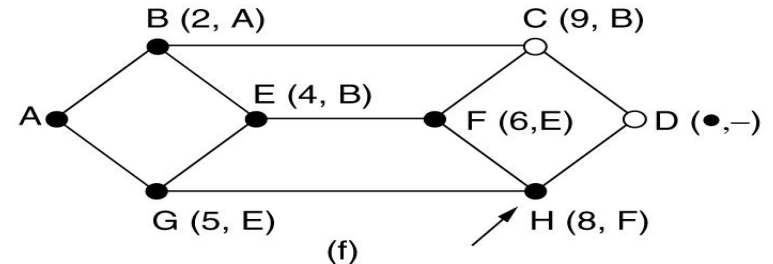
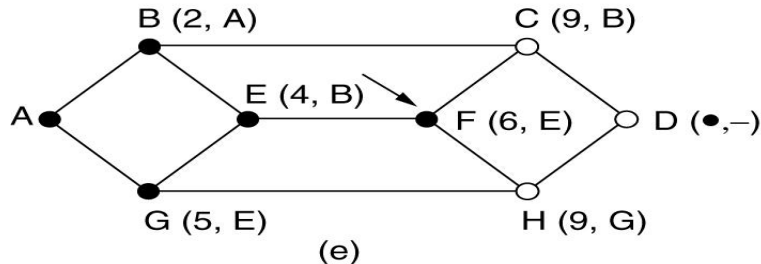
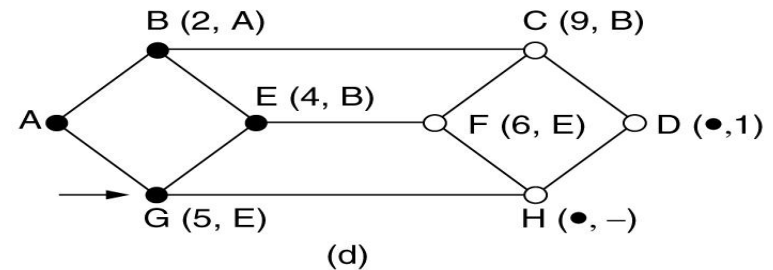
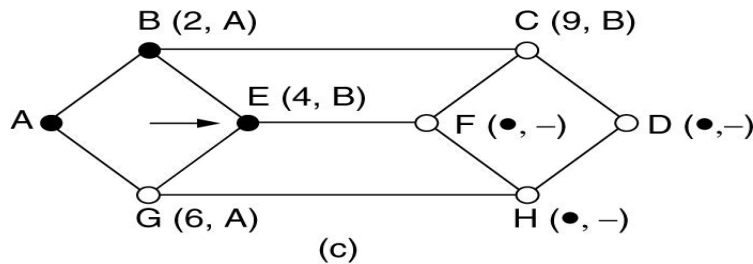
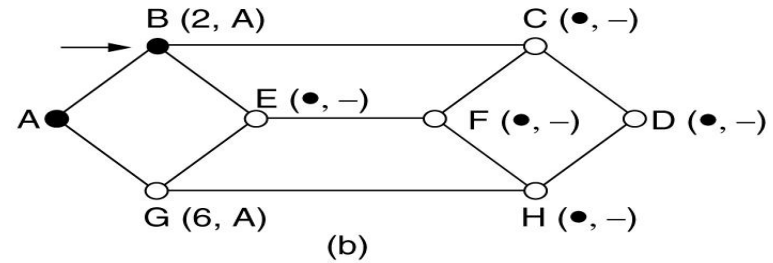
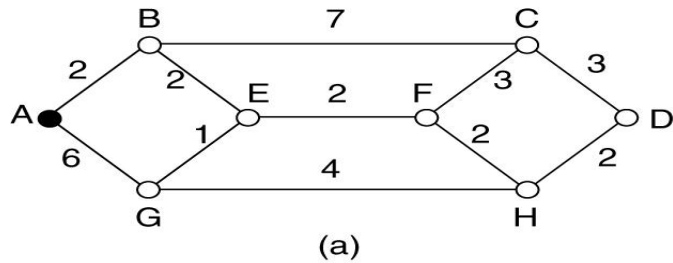
Shortest Path Routing (DIJKSTRA)

Core Concept: A static routing algorithm that finds the shortest path between a source node and all other destinations in a network using a weighted graph.

Main concept

- **Graph Representation:** The network is viewed as a graph where nodes represent routers and edges represent communication links.
- **Metrics for "Length":**
 - **Hops:** Number of edges between nodes.
 - **Geographic Distance:** Physical distance in kilometers.
 - **Delay:** Mean delay measured in milliseconds.
 - **Bandwidth/Cost:** A weighting function combining multiple factors.
- **Key Requirement:** Edge weights (distances) must be **non-negative**.

Shortest Path Routing (DIJKSTRA)



The first 5 steps used in computing the shortest path from A to D.
The arrows indicate the working node.

Shortest Path Routing (DIJKSTRA)

Algorithmic Step-by-Step

1.Mark Source: Set the source node as permanent (filled circle).

2.Relabel Neighbors: Examine nodes adjacent to the working node.

Relabel them with (distance, predecessor) if the new path is shorter than the existing label.

3.Select Minimum: Find the tentatively labeled node with the smallest value in the entire graph.

4.Iterate: Make that node permanent and repeat until the destination node is reached or all nodes are permanent

Shortest Path Routing (DIJKSTRA)

How it Works (The Labeling Method)

The algorithm uses a labeling technique to track the best -known path.

1.Initial State: All nodes are labeled with **infinity** (∞) as no paths are known.

2.Tentative vs. Permanent:

Tentative Label: A path has been found, but a shorter one might still exist.

Permanent Label: The shortest possible path from the source to that node has been discovered.

3.The Working Node: The node with the smallest tentative label is made permanent and becomes the new "working node" to probe its neighbors.

Shortest Path Routing (DIJKSTRA)

Performance & Complexity

- **Time Complexity:** Standard implementation: $O(N^2)$, where N is the number of nodes.
 - With optimized data structures (like Fibonacci Heaps): $O(E + N \log N)$, where E is the number of edges.
- **Space Complexity:** $O(N)$ to store labels and the predecessor list for each node.

Shortest Path Routing (DIJKSTRA)

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Shortest Path Routing (DIJKSTRA)

Advantages

- Optimality:** Guaranteed to find the shortest path in a graph with non-negative weights.
- Efficiency:** Faster than Bellman-Ford for most graphs.
- Global View:** Provides the "ideal" metric against which distributed algorithms can be measured.

Disadvantages

- Requires Global Knowledge:** Every router must have a complete picture of the network topology.
- Static Nature:** Does not naturally handle negative edge weights.
- Resource Intensive:** Storing the entire topology and running calculations can be taxing for very large networks.

Shortest Path Routing (DIJKSTRA)

(b) Consider a network comprising five routers labeled as A, B, C, D, and E. The interconnections and their associated costs are given as follows :

- Link between A and B: Cost 2
- Link between A and C: Cost 4
- Link between B and C: Cost 1
- Link between B and D: Cost 5
- Link between C and D: Cost 3
- Link between C and E: Cost 7
- Link between D and E: Cost 2

Using Dijkstra's algorithm, illustrate the step-by-step process of determining the shortest path from router A to router E. (8)

Need to find: Find the shortest path from **A to E** using the graph provided in the PYQ.

Network Costs:

•A-B: 2 | A-C: 4 | B-C: 1 | B-D: 5 | C-D: 3 | C-E: 7 | D-E: 2

Step	Permanent Node	Tentative Labels (Node, Distance, Predecessor)
1	A (0, -)	B(2, A), C(4, A), D(∞), E(∞)
2	B (2, A)	C(3, B) , D(7, B), E(∞) [Path to C via B is $2+1=3$, which is < 4]
3	C (3, B)	D(6, C) , E(10, C) [Path to D via C is $3+3=6$, which is < 7]
4	D (6, C)	E(8, D) [Path to E via D is $6+2=8$, which is < 10]
5	E (8, D)	Goal reached.

Final Shortest Path: $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E$ (Total Cost: **8**).

Shortest Path Routing (DIJKSTRA)

Q1: What happens if a link fails in a network using Dijkstra's?

A: Because it is a static/centralized algorithm, the entire topology must be redistributed to all nodes, and the algorithm must be re-run to find new paths.

Q2: Why must distances be non-negative?

A: Negative weights can create "negative cycles," which Dijkstra's algorithm cannot handle, as it assumes once a node is permanent, its shortest distance is final.

Q3: How does the algorithm reconstruct the path?

A: Each node is labeled with its predecessor (the node from which it was probed). By tracing these backward from the destination to the source, the path is revealed.

Distance Vector Routing

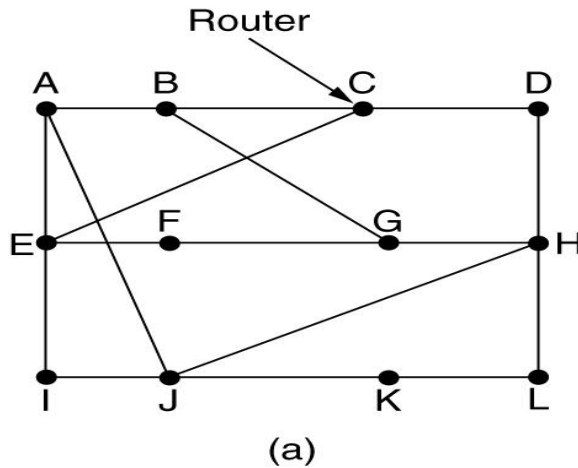
Core Concept: A decentralized, dynamic routing algorithm where routers collaborate to find the shortest path without having a complete map of the network.

Main Concept

Algorithmic Basis: Also known as the **Distributed Bellman-Ford** routing algorithm.

- **Philosophy Behind Working:** "Tell your neighbors about the whole world".
- **The Vector:** Each router maintains a routing table (a vector) containing:
 - **Destination:** Every reachable node in the network.
 - **Cost/Distance:** The best -known distance to that node.
 - **Next Hop:** The outgoing link to use.
- **Historical Implementation:** It was the original ARPANET algorithm and is the foundation for the **Routing Information Protocol (RIP)**.

Distance Vector Routing



To	A	I	H	K	New estimated delay from J	
A	0	24	20	21	8	A
B	12	36	31	28	20	A
C	25	18	19	36	28	I
D	40	27	8	24	20	H
E	14	7	30	22	17	I
F	23	20	19	40	30	I
G	18	31	6	31	18	H
H	17	20	0	19	12	H
I	21	0	14	22	10	I
J	9	11	7	10	0	—
K	24	22	22	0	6	K
L	29	33	9	9	15	K

JA delay is 8	JI delay is 10	JH delay is 12	JK delay is 6	New routing table for J	
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(b)

(a) A subnet. (b) Input from A, I, H, K, and the new routing table for J.

Distance Vector Routing

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Distance Vector Routing

How it Works: The Iterative Process

- **Initial State:** Routers only know the distance to their immediate neighbors (direct links).
- **Information Exchange:** Periodically (every T msec), each router broadcasts its entire routing table to its neighbors.
- **The Calculation:** Upon receiving a vector from neighbor X, router J calculates the distance to destination i as:

$$D_J(i) = \min[D_J(i), \text{cost}(J, X) + D_X(i)]$$

- **Update Rule:** If the new path via X is shorter than the current path, the table is updated with the new cost and the line to X.

Distance Vector Routing

Algorithmic Complexity Analysis

In the context of Distance Vector Routing, complexity is measured by how the algorithm scales with the number of nodes (N) and edges (E):

- **Time Complexity (to Convergence):**

- **Best Case:** $O(D)$, where D is the diameter of the network. Good news propagates at one hop per exchange.

- **Worst Case:** $O(N^2)$ or even $O(\infty)$ during the "Count-to-Infinity" problem.

- **Space Complexity:** $O(N)$ per router. Each router must store an entry for every possible destination in the network.

- **Communication Complexity:** $O(N \times E)$. In each exchange, every router sends a vector of size N to all its neighbors.

Distance Vector Routing

Advantages & Disadvantages

Advantages

- **Low Setup Requirement:** Routers only need to know their immediate neighbors, not the whole topology.
- **Simplicity:** Easier to implement and manage than Link State Routing.
- **Parallel Processing:** It naturally finds the shortest path by exploring all possible paths in parallel.

Disadvantages

- **Slow Convergence:** It takes a long time to settle after a failure.
- **Count-to-Infinity:** It reacts quickly to "good news" but very slowly to "bad news".
- **Routing Loops:** Temporary loops can form during the convergence process.

Distance Vector Routing

The "Count-to-Infinity" Problem

- The Scenario:** If Link A-B fails, B looks at its neighbors for an alternative.
- The Trap:** Neighbor C might claim it has a path to A (not realizing that path went through B).
- The Result:** B and C keep incrementing their distances (3,4,5,...), believing the other has a valid path.
- Solution:** Protocols define a "Maximum Hop Count" (e.g., 16) to represent infinity, forcing the algorithm to stop.

Distance Vector Routing

(b) Consider a network of six routers labelled A, B, C, D, E and F. The cost of the links between the routers is as follows : 5

(i) A-B : 4

(ii) A-E : 5

(iii) B-C : 2

(iv) B-F : 6

(v) C-D : 3

(vi) C-E : 1

(vii) D-F : 7

(viii) F-E : 8

The following information has just arrived at the router C :

- From B : (5, 0, 8, 12, 6, 2)
- From D : (16, 12, 6, 0, 9, 10)
- From E : (7, 6, 3, 9, 0, 4)

The measured delays to B, D and E are 6, 3 and 5 respectively. Using distance vector routing algorithm, give the new routing table for C specifying both the delay and the outgoing line to use.

Problem says: Router C has neighbors B, D, and E with measured delays of 6, 3, and 5 ms respectively. It receives the following delay vectors:

•**From B:** (5, 0, 8, 12, 6, 2)

•**From D:** (16, 12, 6, 0, 9, 10)

•**From E:** (7, 6, 3, 9, 0, 4)

$$D_C(X) = \min\{C(C, B) + D_B(X), C(C, D) + D_D(X), C(C, E) + D_E(X)\}$$

Problem says: Router C has neighbors B, D, and E with measured delays of 6, 3, and 5 ms respectively. It receives the following delay vectors:

- From B:** (5, 0, 8, 12, 6, 2)
- From D:** (16, 12, 6, 0, 9, 10)
- From E:** (7, 6, 3, 9, 0, 4)

To	Via B (Vector + 6)	Via D (Vector + 3)	Via E (Vector + 5)	Best Delay	Line
A	5+6=11	16+3=19	7+5=12	11	B
B	0+6=6	12+3=15	6+5=11	6	B
C	8+6=14	6+3=9	3+5=8	0	-
D	12+6=18	0+3=3	9+5=14	3	D
E	6+6=12	9+3=12	0+5=5	5	E
F	2+6=8	10+3=13	4+5=9	8	B

Distance Vector Routing (2)

A	B	C	D	E	
●	●	●	●	●	Initially
	1	●	●	●	After 1 exchange
	1	2	●	●	After 2 exchanges
	1	2	3	●	After 3 exchanges
	1	2	3	4	After 4 exchanges

(a)

A	B	C	D	E	
●	●	●	●	●	Initially
	1	2	3	4	Initially
	3	2	3	4	After 1 exchange
	3	4	3	4	After 2 exchanges
	5	4	5	4	After 3 exchanges
	5	6	5	6	After 4 exchanges
	7	6	7	6	After 5 exchanges
	7	8	7	8	After 6 exchanges
	⋮				
	●	●	●	●	

(b)

The count-to-infinity problem.

The **Count-to-Infinity Problem** is a major flaw in Distance Vector Routing where the network takes an excessively long time to realize a link has failed because routers keep updating each other with stale information.

1. The "Good News" Propagation (A)

This scenario shows what happens when a router (A) comes online.

•**Initially:** Routers B, C, D, and E do not know A exists (marked by dots/infinity).

•**Step 1:** B learns it is next to A and updates its distance to **1**.

•**Step 2:** In the next exchange, C hears from B that A is 1 hop away. C updates its distance to **2** (1+1).

•**Conclusion:** "Good news" travels fast—exactly one hop per exchange.



(a)

The "Bad News" Problem (B)

This is where the **Count-to-Infinity** occurs. Suppose the link between A and B is suddenly cut.

- Initially:** Everyone has the correct distance to A (B:1, C:2, D:3, E:4).

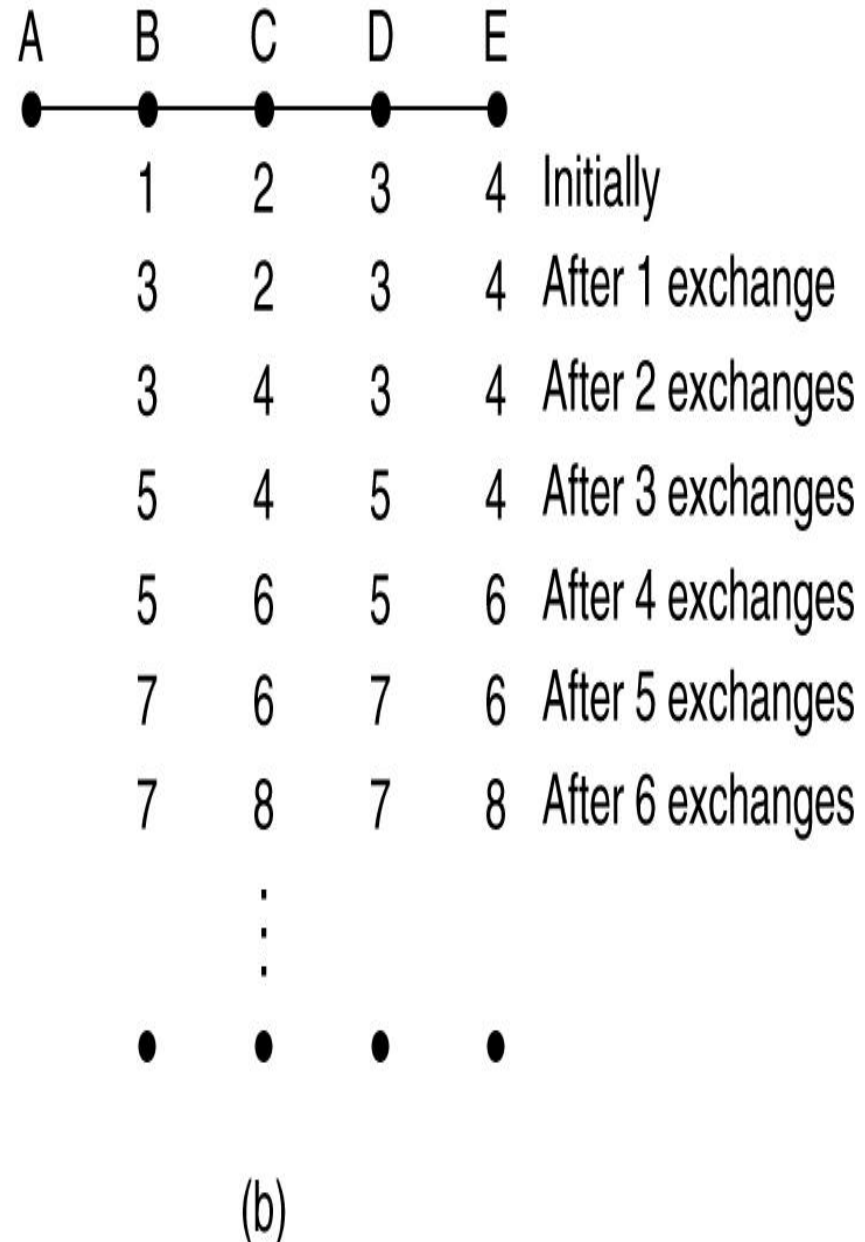
- The Failure:** B can no longer reach A directly.

- The Misunderstanding:** At the first exchange, B hears from C: *"I have a path to A of length 2"*. B doesn't realize C's path actually goes through B itself.

- The Update:** B thinks, *"If C is 2 hops from A, I must be 3 hops from A via C."* B updates its table to 3.

- The Loop:** On the next exchange, C sees B's distance has increased to 3. C thinks its own path must now be 4 (3+1).

- The Result:** The routers continue to "bounce" these updates back and forth, slowly incrementing the distance (5,6,7,8...)



Distance Vector Routing (2)

Feature	Description
Good News	Propagates at a rate of 1 hop per exchange.
Bad News	Propagates "leisurely" (very slowly).
Infinity Value	To limit this problem, "infinity" is usually set to a small number (like 16 in RIP).
Heuristics	Techniques like Split Horizon or Poison Reverse attempt to fix this, but they often fail in complex topologies.

The Count-to-Infinity Problem

Question: The Scenario: Consider a linear network consisting of three routers: **A**, **B**, and **C**. The link costs between **A–B** and **B–C** are both equal to **1**. Initially, the network is fully converged: Router **B** reaches **C** in 1 hop, and Router **A** reaches **C** via **B** with a total distance of 2 hops. Suddenly, the link between **B** and **C** fails. In a Distance Vector Routing environment, explain how the "Count-to-Infinity" problem occurs during the subsequent routing table updates.

The Count-to-Infinity Problem

Answer: the exchange of routing vectors between Router **A** and Router **B** as they attempt to find a new path to Destination **C**.

Initial State:

- $A \rightarrow C = 2$ (via B)
- $B \rightarrow C = 1$

Link B–C fails

Step 1: B tells A: “I can't reach C” $\rightarrow$ distance $= \infty$

Step 2: A still thinks:

$$A \rightarrow C = 2$$

So A tells B: “I can reach C in 2”

Step 3: B updates:

$$B \rightarrow C = 1 + 2 = 3$$

Step 4: A updates:

$$A \rightarrow C = 1 + 3 = 4$$

Loop continues:

$$3 \rightarrow 4 \rightarrow 5 \rightarrow 6 \rightarrow \dots \rightarrow \infty$$

Problem:

Infinite loop (Count-to-Infinity)

Distance Vector Routing (2)

Why is this a problem?

- **Slow Convergence:** The routers will keep counting until they reach a numerical value defined as "infinity".
- **Network Congestion:** While they are counting, they are using bandwidth for useless updates and potentially creating routing loops.
- **The Core Issue:** When a router tells a neighbor it has a path, the neighbor has no way of knowing if it is already part of that path.

Distance Vector Routing

Q: Why was Distance Vector replaced by Link State in ARPANET?

Ans: Because it took too long to converge after topology changes due to the count-to-infinity problem.

Q: What is "Split Horizon"?

Ans: A heuristic to prevent loops where a router does not advertise a route back to the neighbor from which it learned that route.

Q: Which protocol uses DVR today?

Ans: RIP (Routing Information Protocol), which uses hop count as its metric.

Distance Vector Routing

Q1: Justify: "Distance Vector Routing reacts rapidly to good news but leisurely to bad news."

Ans: Good news (like a new link) propagates at one hop per exchange. However, bad news (link failure) causes routers to bounce stale routes off each other, incrementing the distance slowly toward infinity because they lack global knowledge of the path.

Q2: What is the core problem that caused the demise of Distance Vector in ARPANET?

•**Ans:** The algorithm took too long to converge after topology changes due to the count-to-infinity problem.

Q3: Which OSI/TCP-IP layer handles route determination?

Ans: The Network Layer (or Internet Layer in TCP/IP).

A comparison between Distance Vector Routing (DVR) and Dijkstra's Shortest Path Algorithm

1. Topology Visibility

- Dijkstra's:** This algorithm requires a "complete picture" of the network before it can begin. It operates by labeling nodes as "permanent" or "tentative" until the shortest path to all destinations is found.
- Distance Vector:** Routers do not know the full topology. They only maintain a table (vector) of the best known distance to each destination and which link to use to get there. They learn about the rest of the network purely through exchanges with neighbors

A comparison between Distance Vector Routing (DVR) and Dijkstra's Shortest Path Algorithm

2. Performance and "Bad News"

- Distance Vector:** This method reacts rapidly to "good news" (new links) but very "leisurely to bad news" (link failures). This is because routers may keep using stale information from each other, leading to the **Count-to-Infinity problem**.
- Dijkstra's:** Because it relies on a global map, if a link fails, the topology is updated, and the algorithm simply re-calculates the new shortest path without the "counting" loops seen in DVR

A comparison between Distance Vector Routing (DVR) and Dijkstra's Shortest Path Algorithm

3. Real-World Protocol Usage

- **Distance Vector:** Historically used in the **ARPANET** and currently used in the Internet under the name **RIP (Routing Information Protocol)**.
- **Dijkstra's:** Serves as the foundation for **Link State Routing** protocols, such as **OSPF (Open Shortest Path First)** and **IS-IS**, which replaced Distance Vector in large, modern networks.

The Network Layer in the Internet

- The IP Protocol
- IP Addresses
- Internet Control Protocols
- OSPF: Open Shortest Path First – The Interior Gateway Routing Protocol
- BGP: Border Gateway Protocol – The Exterior Gateway Routing Protocol
- Internet Multicasting
- Mobile IP
- IPv6

Design Principles for Internet

1. **Make sure it works:** Before finalizing any standard or design, multiple prototypes should be built to verify that they can successfully communicate with each other.
2. **Keep it simple:** When in doubt, always choose the simplest solution.
3. **Make clear choices:** If there are multiple ways to achieve the exact same thing, choose just one .
4. **Exploit modularity:** The network should be built using independent protocol stacks (like the TCP/IP layers).

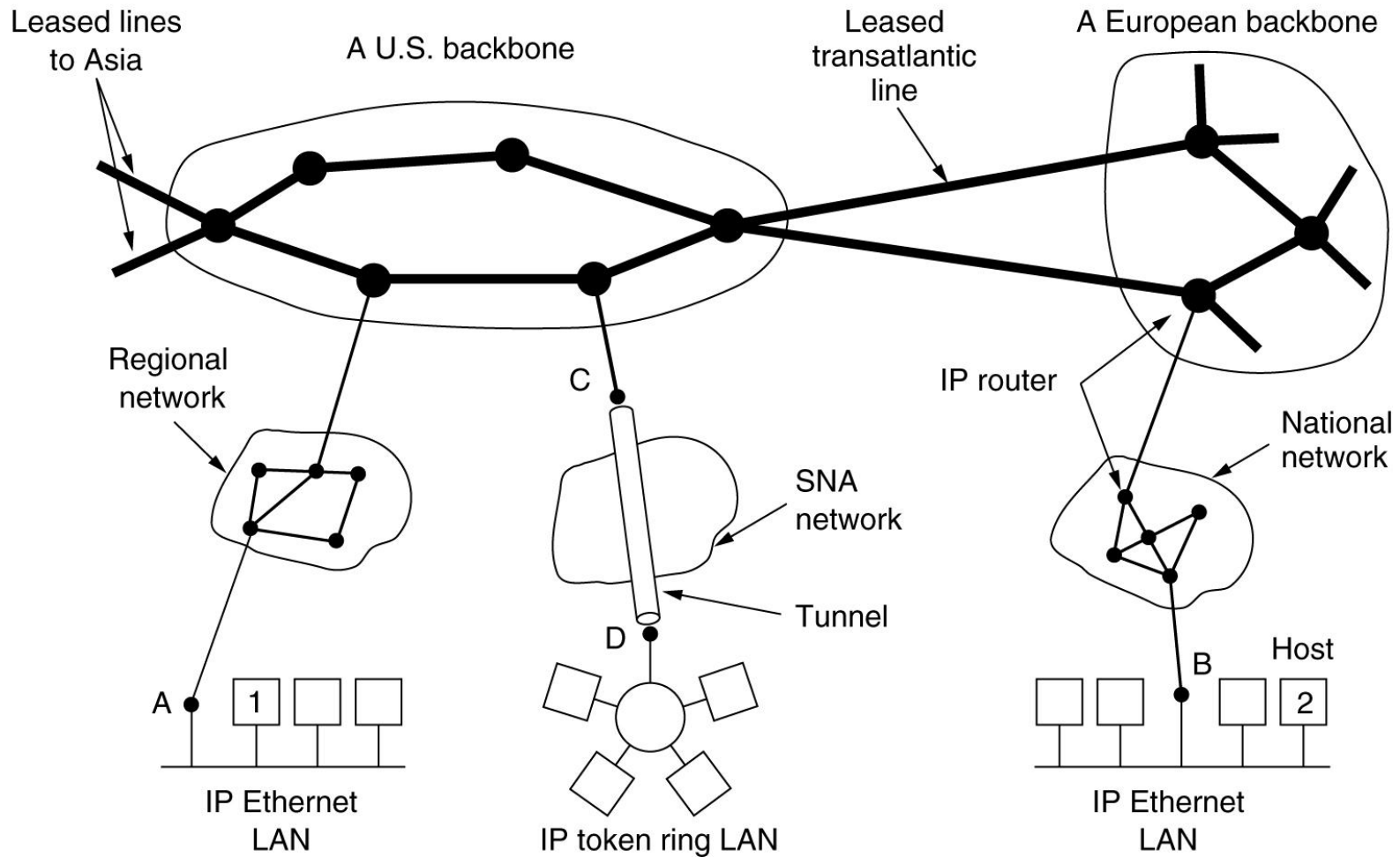
Design Principles for Internet

- 5. **Expect heterogeneity:** The network's core design must remain simple, general, and flexible enough to accommodate this constant diversity.
- 6. **Avoid static options and parameters:** Hardcoded fixed choices should be avoided whenever possible..
- 7. **Look for a good design:** Don't ruin a solid design just to accommodate extreme edge cases.
- 8. **Be strict when sending and tolerant when receiving:** A host should only send out packets that rigorously adhere to the official standards.

Design Principles for Internet

9. **Think about scalability:** The design must be able to support millions of hosts and billions of users without failing.
10. **Consider performance and cost:** The system must be fast and economically viable.

Collection of Subnetworks



The Internet is an interconnected collection of many networks.

IP address

- a) An IP address is an identifier for a particular machine on a particular network. It is part of a scheme to identify computers on the internet.
- b) The network portion of the IP address is allocated to internet service provider(ISP) by the InterNIC, under authority of the internet assigned number authority(IANA).
- c) ISPs then assign the host portion of the IP address to the machines on the networks that they operate.
- d) Which section of the IP address represent the network & which sections represent the machine will depend on what “class” of IP address is assigned to a network.

IPv4

- It is 32 bit number represented in 4 decimal number where each decimal number is of 8 bit (octet) is separated by a dot(.).
- Thus representation known as doted decimal representation.
- IP address consists of 2 components the network id and the host id.

Network id:-it is the number assigned to a network in the internet.

Host id:- it represents the id assigned to a host in the network.

IPv4

- a) Network id:-it is the number assigned to a network in the internet.

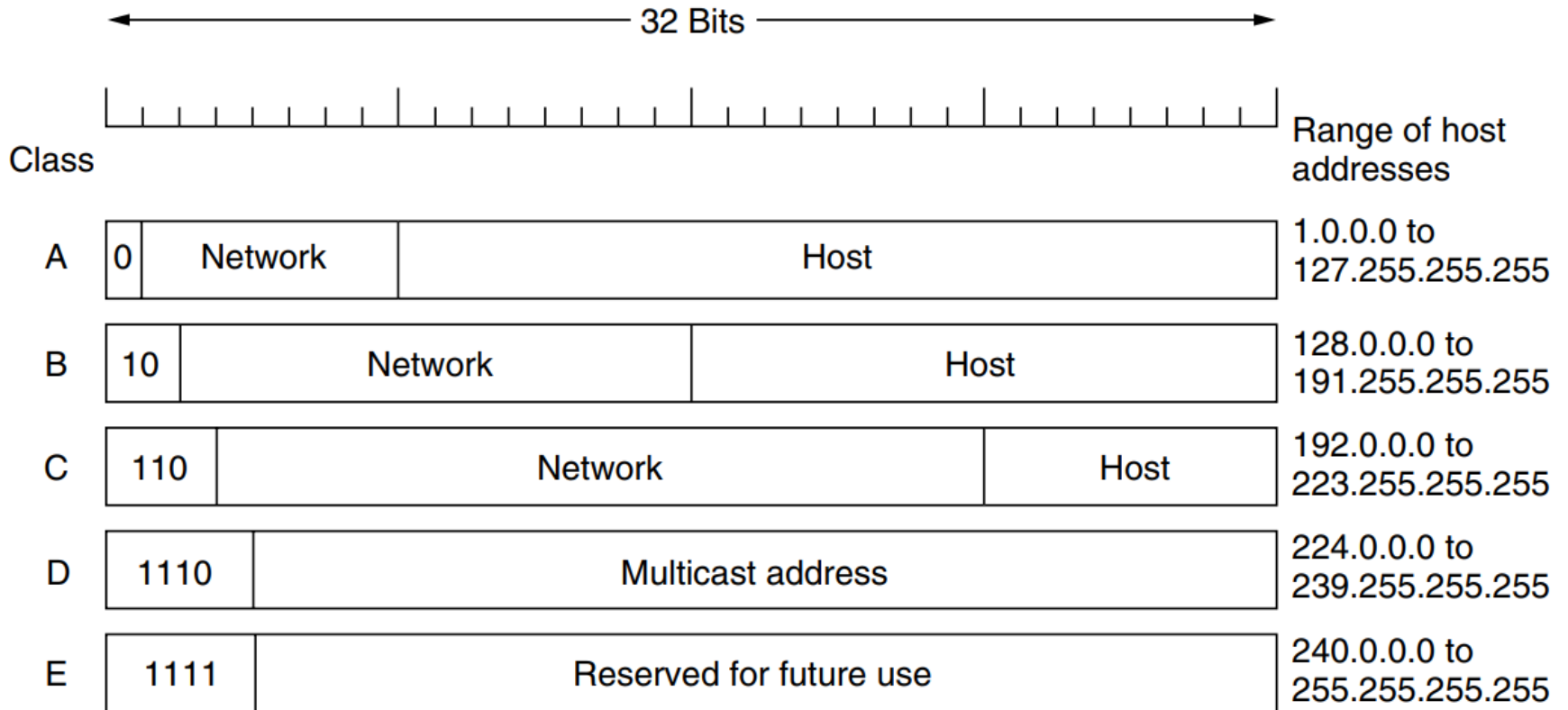
Host id:- it represents the id assigned to a host in the network.

- a) IPv4 allows 2^{32} (4294967296) unique address which section of IP address show the network id & which section show the machine or host id depend on the class network.

IPv4 cont...

- a) There are five classes of IP addresses:-
- Class A : 8-bit network, 24-bit host (starts with 0). Used for massive networks.
 - Class B : 16-bit network, 16-bit host (starts with 10). "The Three Bears Problem" – too big for most, too small for massive ISPs.
 - Class C: 24-bit network, 8-bit host (starts with 110). Supports up to 256 hosts.
 - Class D: Multicast addresses (starts with 1110).
 - Class E

IPv4 cont...



IPv4 cont...

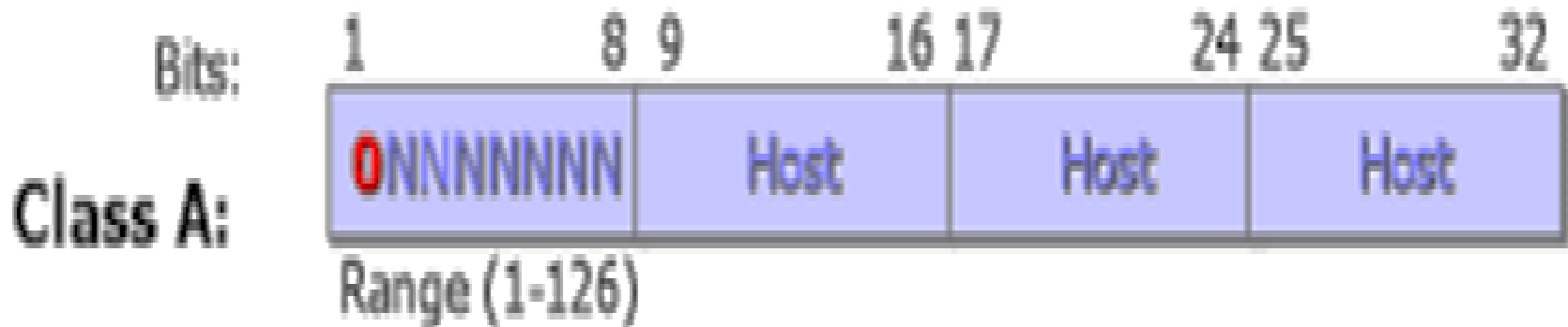
Class A:-

- The first byte is a network id (8 bits) & the last 3 bytes are for host id (24 bits).
- The first bit is '0'.
- Range of network number-1.0.0.0 to 127.0.0.0
- Number of possible networks-127(1-126 usable, 127 is reserved)

❖ IPv4 cont...

Class A:-

- Number of possible values in the host portion- $(2^{24}):16,777,216$
- It is used for large network.



❖ IPv4 cont...

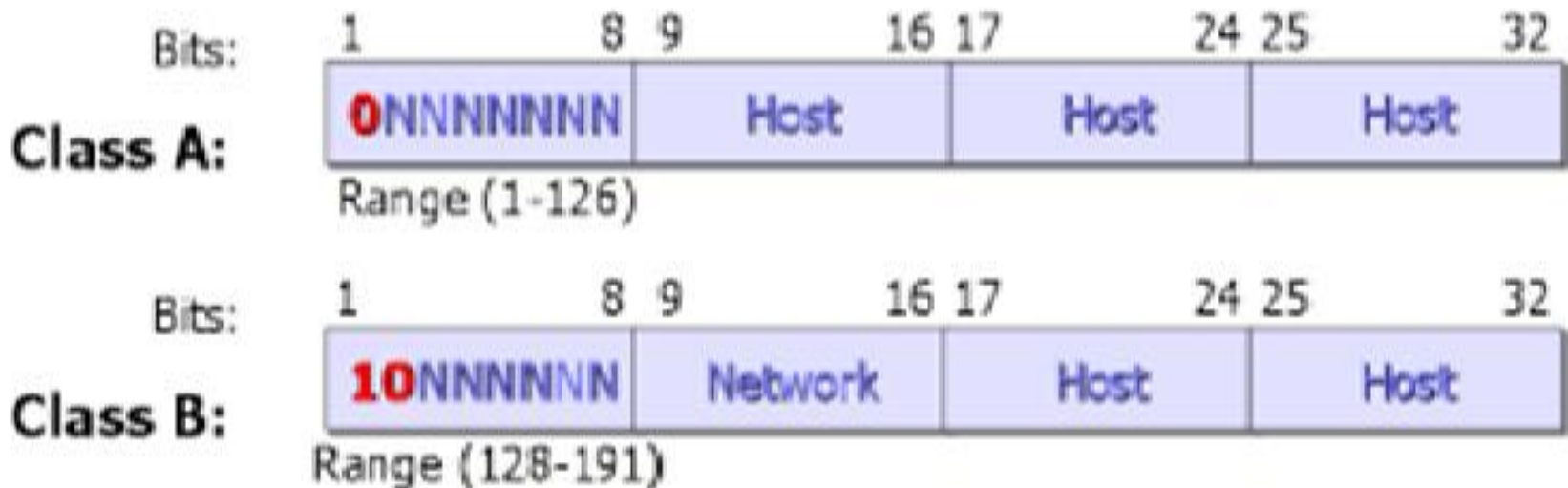
Class B:-

- The first 2 bytes are a network id (16 bits) & the last 2 bytes are for host id (16 bits).
- The first 2 bits are '10'.
- Range of network number- 128.0.0.0 to 191.255.0.0
- Number of possible networks- $(2^{16})16,384$

❖ IPv4 cont...

Class B:-

- Number of possible values in the host portion- 65536
- Used for medium size network.



❖ IPv4 cont...

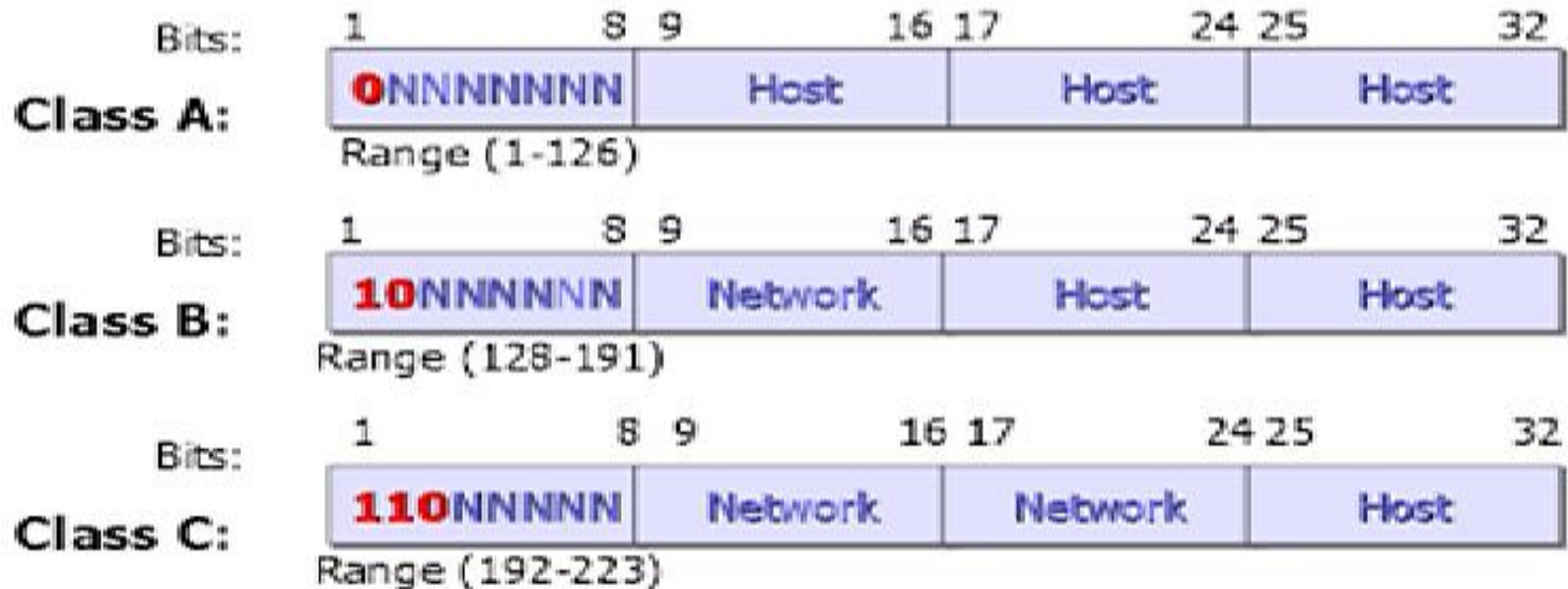
Class C:-

- The first 3 bytes are a network id (24 bits) & the last 1 byte are for host id (8 bit).
- The first 3 bits are '110'.
- Range of network number- 192.0.0.0 to 223.255.255.0
- Number of possible networks- 2,097,152

❖ IPv4 cont...

Class C:-

- Number of possible values in the host portion- 256
- Used in local area network(LAN).



❖ IPv4 cont...

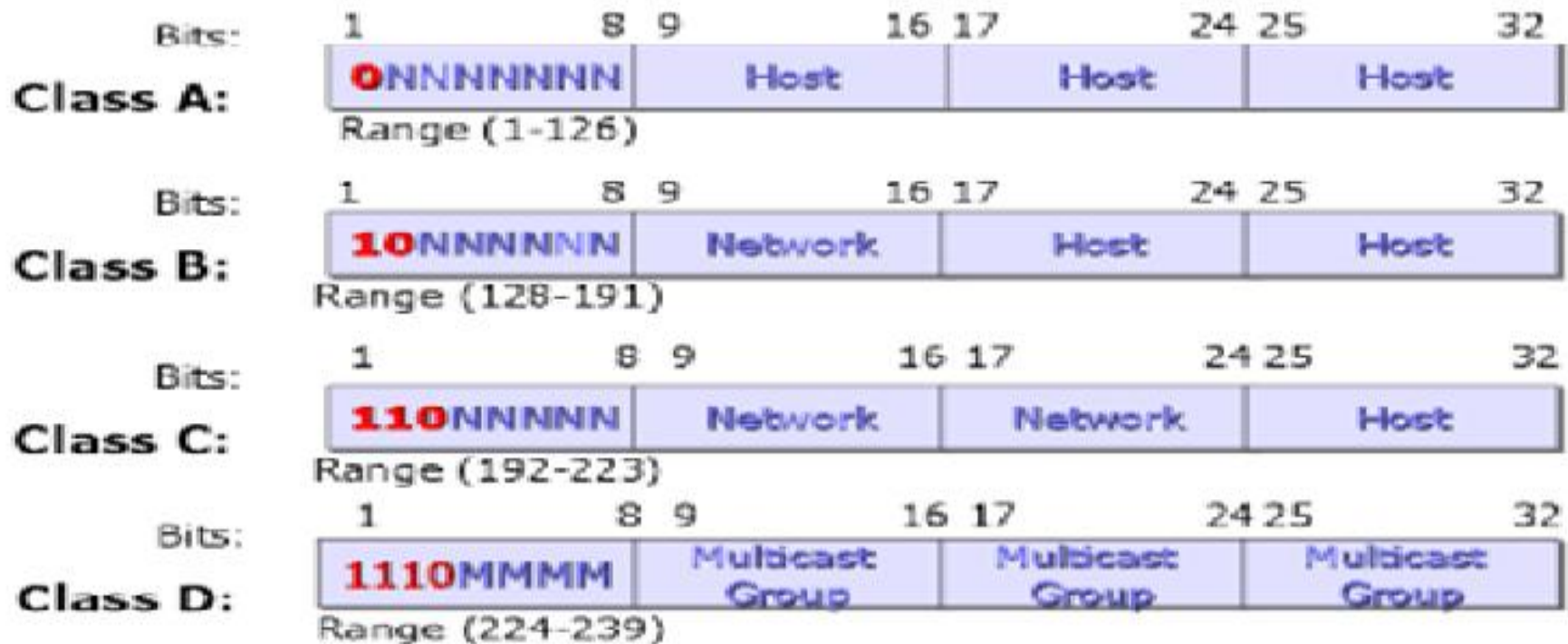
Class D:-

- An IP address which belong to class D has the first octet has its 4bit set to '1110'.
- Range of network number- 224.0.0.0 to 239.255.255.255

❖ IPv4 cont...

Class D:-

- Used for multicasting



❖ IPv4 cont...

Class E:-

- It reserved for experimental & for future testing purpose.
- Range of network number- 240.0.0.0 to 255.255.255.255

Limitations and Problems of IPv4

- Root Cause:** IPv4 uses 32-bit addresses, allowing for only about 4.3 billion unique IP addresses.
- The Reality:** With the explosion of smartphones, home networks, and IoT devices, the Internet is literally running out of IPv4 addresses.
- Inefficient Allocation (Classful Routing):** The original class-based system wasted millions of addresses. For example, a Class B network was assigned 65,536 addresses, which was far too many for most organizations but was assigned anyway because Class C (256 addresses) was too small (the "Three Bears Problem").

Network Addressing

IP addresses identify both the network and the host

❑ **Subnetting:** Splitting a single large network block into smaller, manageable internal pieces (subnets). It masks internal network complexity from the outside world.

Subnet mask

–Indicates how much of the IP address represents the network or subnet

Standard (default) subnet masks:

- Class A**, subnet mask is 255.0.0.0
- Class B**, subnet mask is 255.255.0.0
- Class C**, subnet mask is 255.255.255.0

Network Addressing Cont...

- TCP/IP hosts use the combination of the IP address and the subnet mask
 - To determine if other addresses are local or remote
 - The binary AND operation is used to perform the calculation

Subnetting

- –Manipulation of the subnet mask to get more network numbers

Network Addressing cont...

- **Subnet address**
 - Network is identified by the first, or first few, octets
 - A TCP/IP host must have a nonzero host identifier
- **Broadcast address**
 - When the entire host portion of an IP address is all binary ones
 - Examples: 190.55.255.255 and 199.192.65.63

Network Addressing cont...

Subnet ID:	199.192.65.0	11000111.11000000.01000001.00000000
Subnet mask:	255.255.255.0	11111111.11111111.11111111.00000000
Broadcast Address:	199.192.65.255	11000111.11000000.01000001.11111111

Broadcast addresses

Subnet ID:	199.192.65.32	11000111.11000000.01000001.00100000
Subnet mask:	255.255.255.224	11111111.11111111.11111111.11100000
Broadcast Address:	199.192.65.63	11000111.11000000.01000001.00111111

Broadcasts on partially octets

Broadcast Types

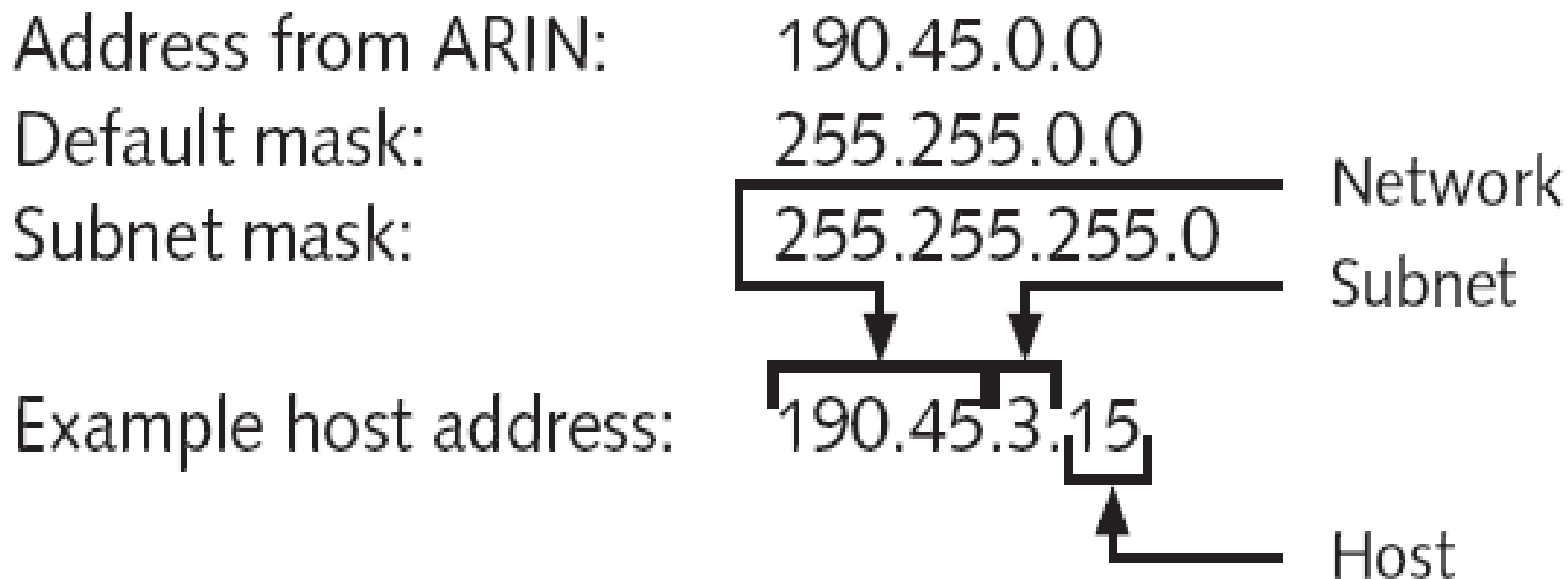
- **Flooded broadcasts**
 - Broadcasts for any subnet
 - Use the IP address 255.255.255.255
 - A router does not propagate flooded broadcasts because they are considered local
- **Directed broadcasts** are for a specific subnet
 - Routers can forward directed broadcasts
 - For example, a packet sent to the Class B address 129.30.255.255 would be a broadcast for network 129.30.0.0

Subdividing IP Classes

Reasons for subnetting

- To match the physical layout of the organization
- –To match the administrative structure of the organization
- To plan for future growth
- To reduce network traffic

Subdividing IP Classes cont...



Dividing a Class B network

Subnet Masking

- When network administrators create subnets
 - They borrow bits from the original host field to make a set of subnetworks
 - The number of borrowed bits determines how many subnetworks and hosts will be available
- Class C addresses also can be subdivided
 - Not as many options or available masks exist because only the last octet can be manipulated with this class

Subnet Mask	Subnets on Network	Hosts per Subnet
255.255.128.0	2	32,766
255.255.192.0	4	16,382
255.255.224.0	8	8,190
255.255.240.0	16	4,094
255.255.248.0	32	2,046
255.255.252.0	64	1,022
255.255.254.0	128	510
255.255.255.0	256	254
255.255.255.128	512	126
255.255.255.192	1,024	62
255.255.255.224	2,048	30
255.255.255.240	4,096	14
255.255.255.248	8,192	6
255.255.255.252	16,384	2

Subnet Masking cont...

Subnet Mask	Subnets on Network	Hosts per Subnet
255.255.255.128	2	126
255.255.255.192	4	62
255.255.255.224	8	30
255.255.255.240	16	14
255.255.255.248	32	6
255.255.255.252	64	2

Class C subnet mask

Subnet Masking cont...

Binary Place Values	128	64	32	16	8	4	2	1	Decimal Equivalents
	1	0	0	0	0	0	0	0	=128
	1	1	0	0	0	0	0	0	=192
	1	1	1	0	0	0	0	0	=224
Binary Digits	1	1	1	1	0	0	0	0	=240
	1	1	1	1	1	0	0	0	=248
	1	1	1	1	1	1	0	0	=252
	1	1	1	1	1	1	1	0	=254
	1	1	1	1	1	1	1	1	=255
	Subnet mask values								

Learning to Subnet

- a) Suppose you had a network with:
 - ⌘ Five different segments
 - ⌘ Somewhere between 15 and 20 TCP/IP hosts on each network segment
- a) You just received your Class C address from ARIN (199.1.10.0)
- b) Only one subnet mask can handle your network configuration: 255.255.255.224
 - ⌘ This subnet mask will allow you to create eight subnetworks and to place up to 30 hosts per network

Learning to Subnet cont...

- Determine the subnet identifiers (IP addresses)
 - Write the last masking octet as a binary number
 - Determine the binary place of the last masking digit
- Calculate the subnets
 - Begin with the major network number (subnet zero) and increment by 32
 - Stop counting when you reach the value of the mask
- Determine the valid ranges for your hosts on each subnet
 - Take the ranges between each subnet identifier
 - Remove the broadcast address for each subnet

Learning to Subnet cont...

Class C Address: 199.1.10.0
Standard Mask: 255.255.255.0
Selected Mask: 255.255.255.224

	128	64	32	16	8	4	2	1
224	1	1	1	0	0	0	0	0

Subnet masking example

Learning to Subnet cont...

Subnet Identifier	Valid Host Range	Broadcast Address for Subnet
199.1.10.0	199.1.10.1 – 199.1.10.30	199.1.10.31
199.1.10.32	199.1.10.33 – 199.1.10.62	199.1.10.63
199.1.10.64	199.1.10.65 – 199.1.10.94	199.1.10.95
199.1.10.96	199.1.10.97 – 199.1.10.126	199.1.10.127
199.1.10.128	199.1.10.129 – 199.1.10.158	199.1.10.159
199.1.10.160	199.1.10.161 – 199.1.10.190	199.1.10.191
199.1.10.192	199.1.10.193 – 199.1.10.222	199.1.10.223
199.1.10.224	199.1.10.225 – 199.1.10.254	199.1.10.255

Learning to Subnet cont...

Binary			Binary		
Decimal Mask	0 224	00000000 11100000	Decimal Mask	32 224	00100000 11100000
Decimal Mask	64 224	01000000 11100000	Decimal Mask	96 224	01100000 11100000
Decimal Mask	128 224	10000000 11100000	Decimal Mask	160 224	10100000 11100000
Decimal Mask	192 224	11000000 11100000	Decimal Mask	224 224	11100000 11100000

Subnetting Formulas

- Consider memorizing the following two formulas:
 $2^y = \#$ of usable subnets (where y is the number of bits borrowed)

 $2^x - 2 = \#$ of usable hosts per subnet (where x is the number of bits remaining in the host field after borrowing)

Subnetting Formulas cont...

C Address	199.4.10.0	11000111.11000000.01000001.00000000
Standard mask	255.255.255.0	11111111.11111111.11111111.00000000
Mask	255.255.255.240	11111111.11111111.11111111.11110000

y = 4 (borrowed bits)

x = 4 (bits left in host field after borrowing)

Formulas:

2^y = # of usable subnets

$2^x - 2$ = # of usable hosts per subnet

2^4 = 16 usable subnets

$2^4 - 2$ = 14 usable hosts per subnet

Subnetting Formulas cont...

	128	64	32	16	8	4	2	1
240	1	1	1	1	0	0	0	0

Below is a list of the last octets for the 16 subnets created from network number 199.4.10.0 with the subnet mask 255.255.255.240

0	128
16	144
32	160
48	176
64	192
80	208
96	224
112	240

Subnetwork numbers will increment by 16, as it is the decimal equivalent of the right-most significant digit in the mask

Classless Inter-Domain Routing (CIDR)

- **Classless Inter-Domain Routing (CIDR)**
 - Developed to slow the exhaustion of IP addresses
 - Based on assigning IP addresses on criteria other than octet boundaries
- CIDR addressing method allows the use of a **prefix** to designate the number of network bits in the mask
 - Example: 200.16.1.48 /25 (CIDR notation)
 - The first 25 bits in the mask are network bits (1s)
- The prefix can be longer than the default subnet mask (subnetting) or it can be shorter than the default mask (**supernetting**)

Supernet Mask

- a) A supernet mask is the reverse of a subnet mask. A subnet mask for class C has more 1s than the default mask for this class. A supernet mask for class C has less 1s than the default mask for this class.

Supernetting (Summarization)

Summarization

- Also known as route aggregation or **supernetting**
- Allows many IP subnets to be advertised as one
 - Reduces the number of entries in the router's routing table

Summarize a group of subnets

- Count the number of bits that are common to all of the networks you want to advertise
- Then use the prefix that identifies the number of common bits

Summarization cont...

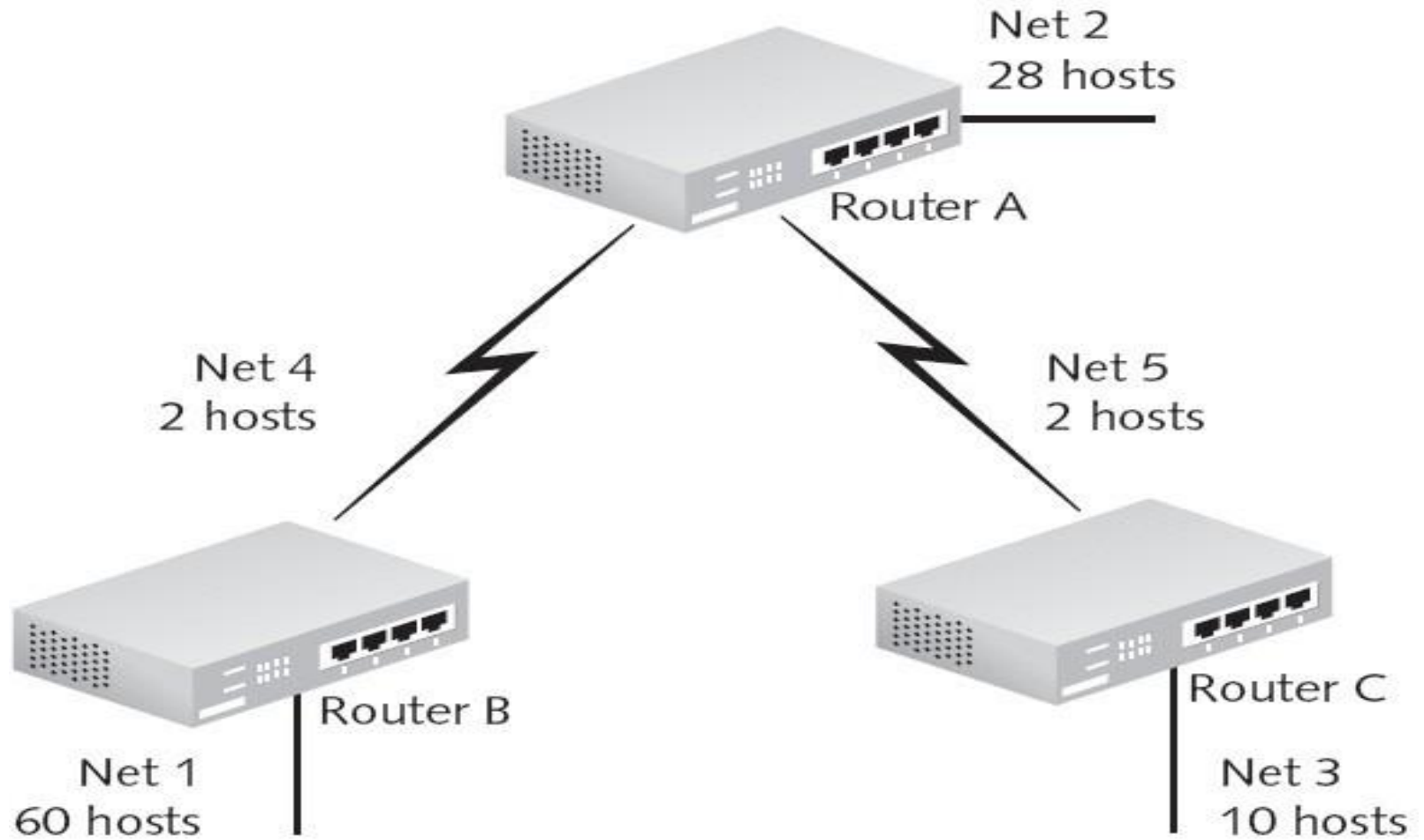
Decimal	Binary Equivalent
213.64.132.0 /24	11010101.01000000.10000100.00000000
213.64.133.0 /24	11010101.01000000.10000101.00000000
213.64.134.0 /24	11010101.01000000.10000110.00000000
213.64.135.0 /24	11010101.01000000.10000111.00000000

Example of summarization

Variable Length Subnet Masks

- **Variable length subnet masking (VLSM)**
 - Allows different masks on the subnets
 - Essentially done by subnetting the subnets
- Basic routing protocols such as RIP version 1 and IGRP
 - Do not support VLSM because they do not carry subnet mask information in their routing table updates
 - Are classful routing protocols
- RIP version 2, OSPF, or EIGRP are classless protocols

Example of internetwork for VLSM (Variable Length Subnet Mask)



Variable Length Subnet Masks

cont...

192.168.59.128 /30

192.168.59.160 /30

192.168.59.132 /30

192.168.59.164 /30

192.168.59.136 /30

192.168.59.168 /30

192.168.59.140 /30

192.168.59.172 /30

192.168.59.144 /30

192.168.59.176 /30

192.168.59.148 /30

192.168.59.180 /30

192.168.59.152 /30

192.168.59.184 /30

192.168.59.156 /30

192.168.59.188 /30

VLSM IP scheme created from 192.168.59.128/26

Variable Length Subnet Masks

cont...

Major Network	Original Subnets	Subnetted Subnets Using VLSM	Subnet Assignments
192.168.59.0 /24	192.168.59.0 /26		Net 1
	192.168.59.64 /26	192.168.59.64 /27	Net 2
		192.168.59.96 /27	Net 3
	192.168.59.128 /26	192.168.59.128 /30	Net 4
		192.168.59.132 /30	Net 5
		192.168.59.136 through 192.168.59.188	Reserved
	192.168.59.192 /26		Reserved

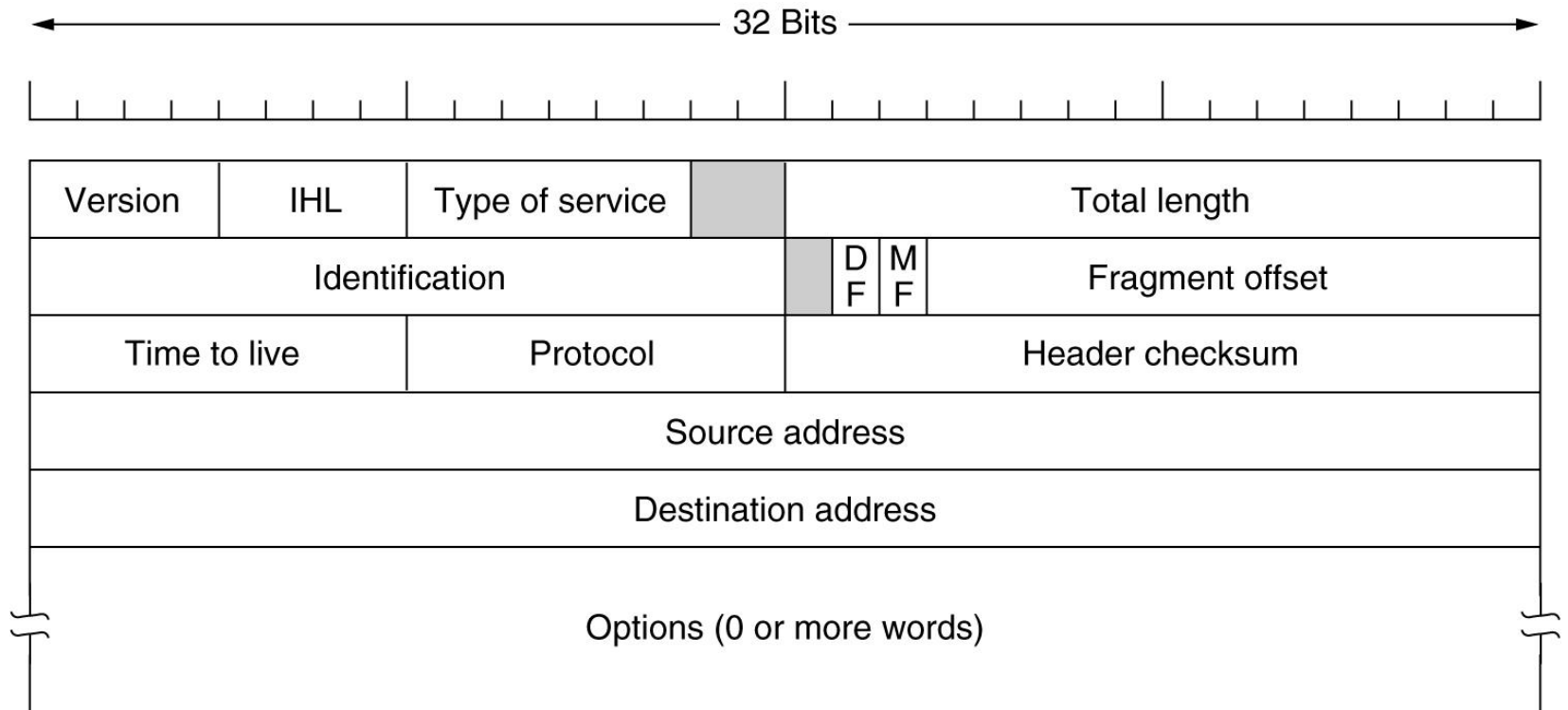
VLSM IP scheme for 192.168.59.0

Working with Hexadecimal Numbers

- **Hexadecimal** numbering system is base 16
 - 16 numerals are used to express any given number
 - Numerals include 0 through 9 as well as A through F
 - For example, the decimal number 192 is C0 in hexadecimal
- Often you will come across hexadecimal numbers when working with computers and networking
 - The MAC address is a 12-digit hexadecimal number
- Computers typically process information in 8-bit chunks (bytes)
 - Easier to express bytes with two hex digits

Binary	Hexadecimal	Decimal
0000	0	0
0001	1	1
0010	2	2
0011	3	3
0100	4	4
0101	5	5
0110	6	6
0111	7	7
1000	8	8
1001	9	9
1010	A	10
1011	B	11
1100	C	12
1101	D	13
1110	E	14
1111	F	15

The IP Protocol (IPv4)



The IPv4 (Internet Protocol) header.

The IP Protocol (2)

Option	Description
Security	Specifies how secret the datagram is
Strict source routing	Gives the complete path to be followed
Loose source routing	Gives a list of routers not to be missed
Record route	Makes each router append its IP address
Timestamp	Makes each router append its address and timestamp

Some of the IP options.

IP Addresses

	← 32 Bits →																																	
Class																																	Range of host addresses	
A	0	Network																Host																1.0.0.0 to 127.255.255.255
B	10	Network														Host																		128.0.0.0 to 191.255.255.255
C	110	Network																				Host												192.0.0.0 to 223.255.255.255
D	1110	Multicast address																																224.0.0.0 to 239.255.255.255
E	1111	Reserved for future use																																240.0.0.0 to 255.255.255.255

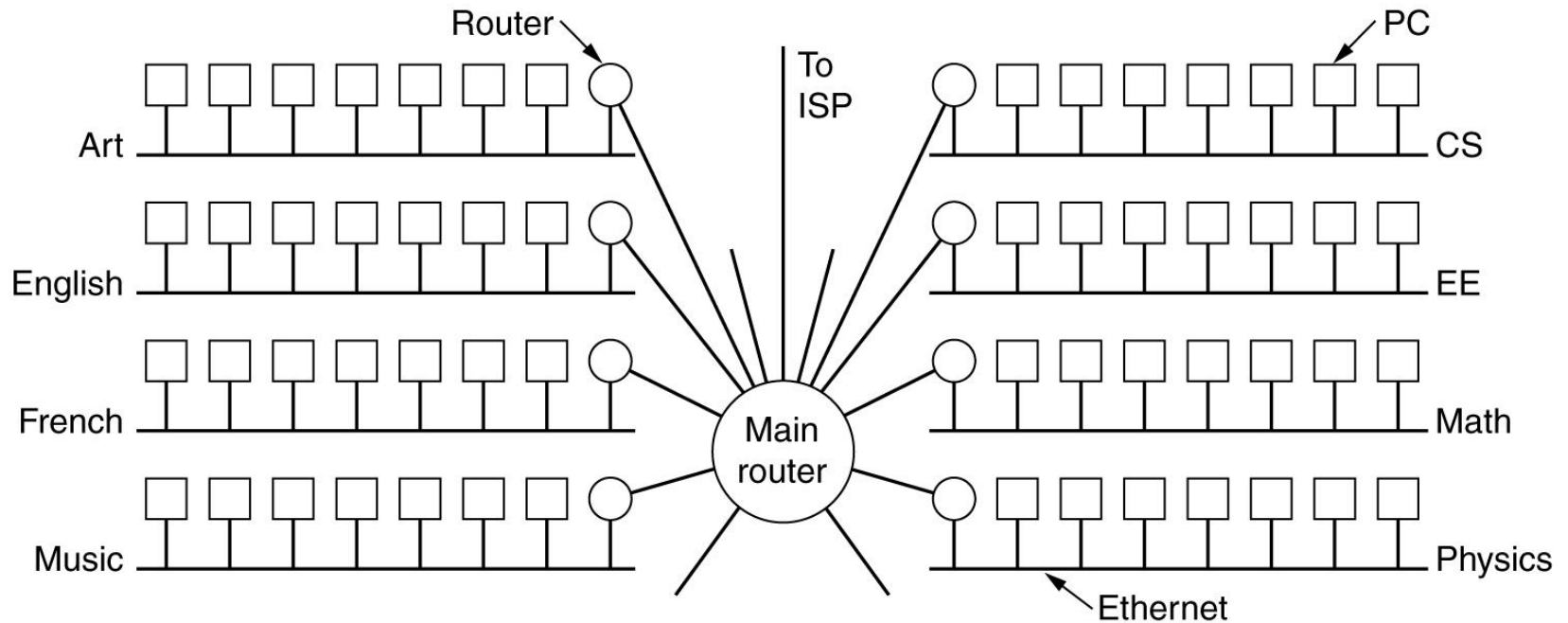
IP address formats.

IP Addresses (2)

0 0																														This host										
0 0										...										0 0										Host	A host on this network									
1 1																														Broadcast on the local network										
Network										1 1 1 1										...										1 1 1 1										Broadcast on a distant network
127					(Anything)																									Loopback										

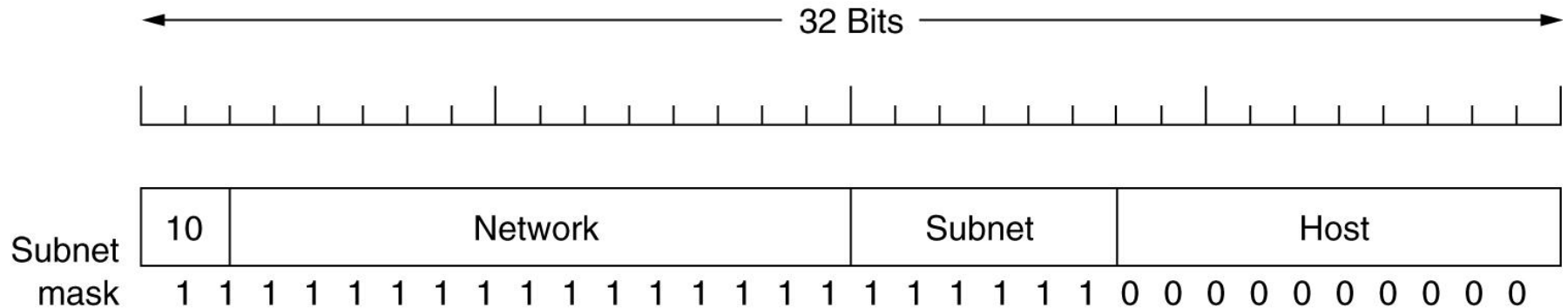
Special IP addresses.

Subnets



A campus network consisting of LANs for various departments.

Subnets (2)



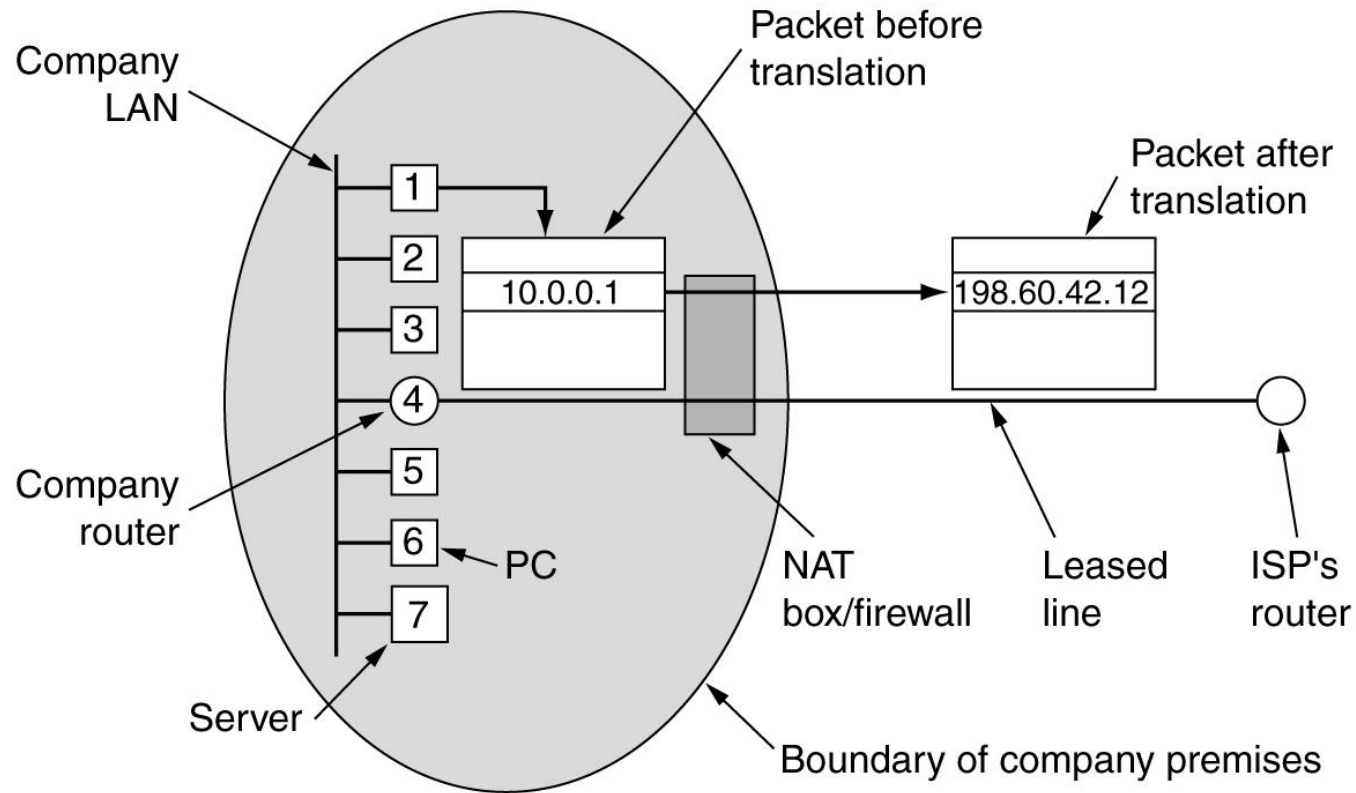
A class B network subnetted into 64 subnets.

CDR – Classless InterDomain Routing

University	First address	Last address	How many	Written as
Cambridge	194.24.0.0	194.24.7.255	2048	194.24.0.0/21
Edinburgh	194.24.8.0	194.24.11.255	1024	194.24.8.0/22
(Available)	194.24.12.0	194.24.15.255	1024	194.24.12/22
Oxford	194.24.16.0	194.24.31.255	4096	194.24.16.0/20

A set of IP address assignments.

NAT – Network Address Translation



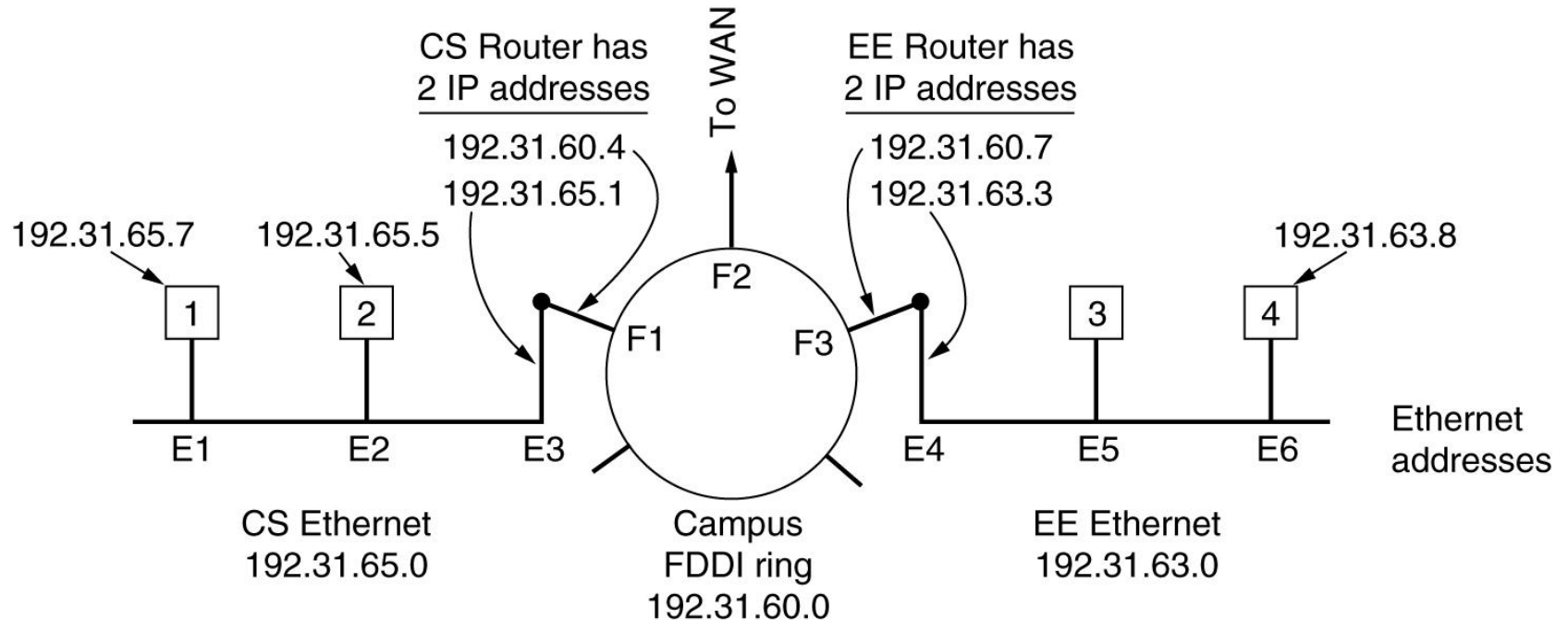
Placement and operation of a NAT box.

Internet Control Message Protocol

Message type	Description
Destination unreachable	Packet could not be delivered
Time exceeded	Time to live field hit 0
Parameter problem	Invalid header field
Source quench	Choke packet
Redirect	Teach a router about geography
Echo request	Ask a machine if it is alive
Echo reply	Yes, I am alive
Timestamp request	Same as Echo request, but with timestamp
Timestamp reply	Same as Echo reply, but with timestamp

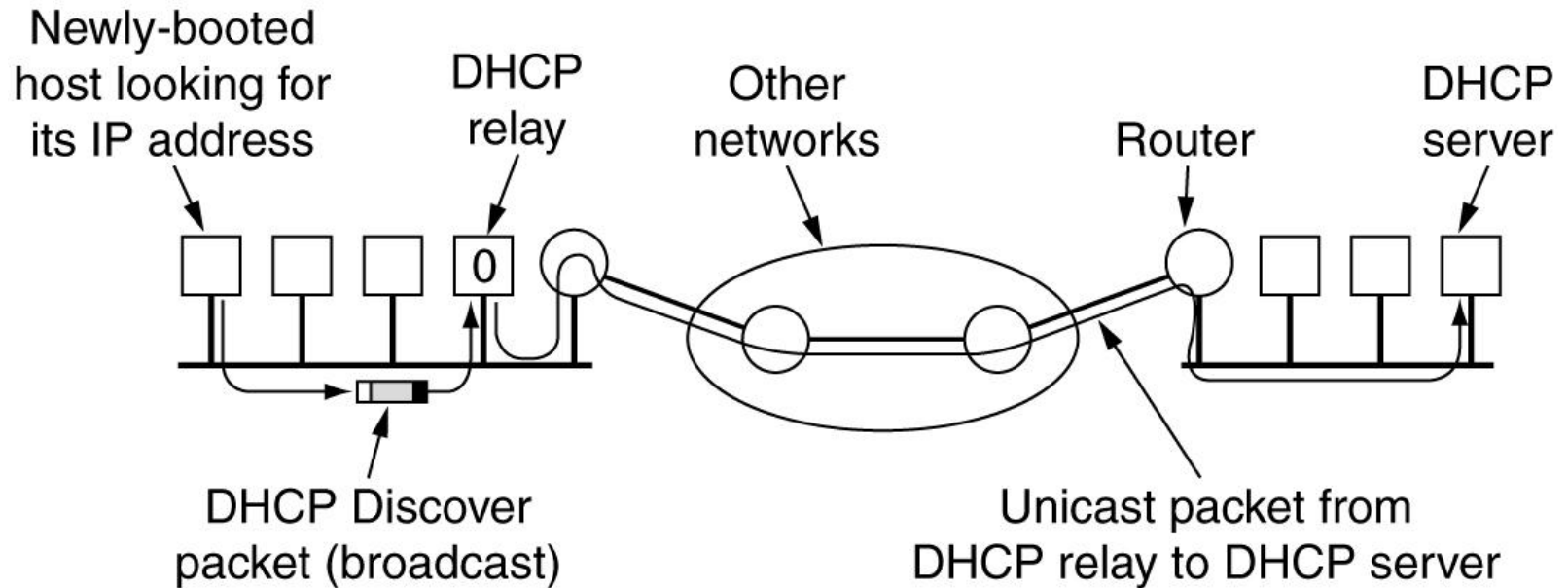
The principal ICMP message types.

ARP– The Address Resolution Protocol



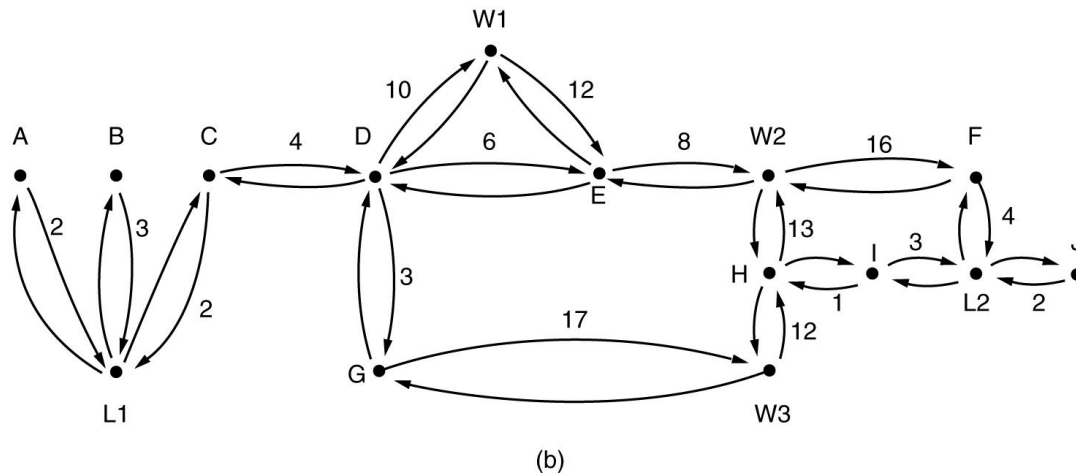
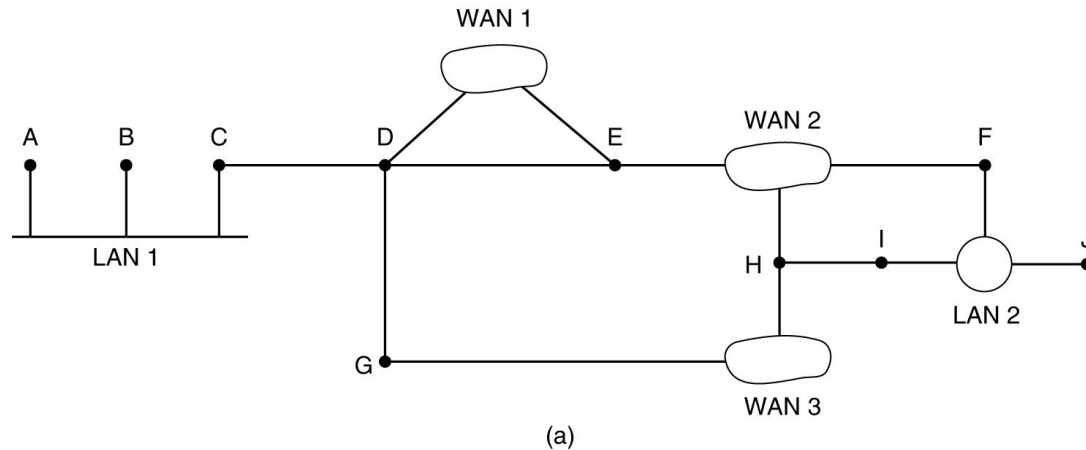
Three interconnected /24 networks: two Ethernet and an FDDI ring.

Dynamic Host Configuration Protocol



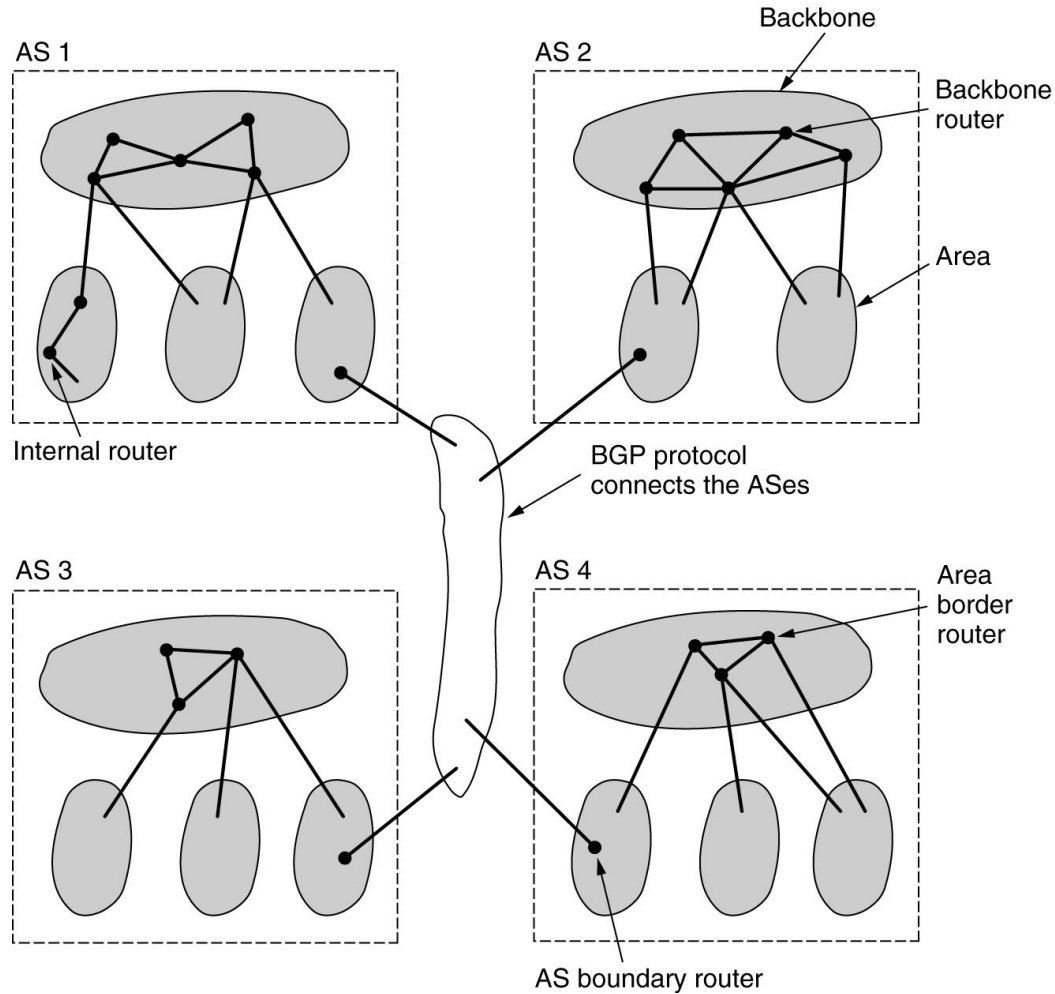
Operation of DHCP.

OSPF – The Interior Gateway Routing Protocol



(a) An autonomous system. (b) A graph representation of (a).

OSPF (2)



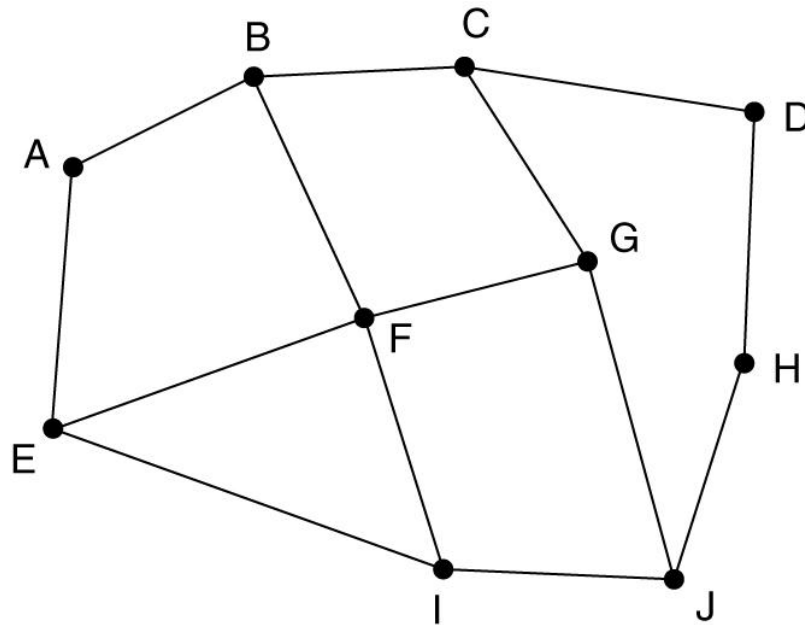
The relation between ASes, backbones, and areas in OSPF.

OSPF (3)

Message type	Description
Hello	Used to discover who the neighbors are
Link state update	Provides the sender's costs to its neighbors
Link state ack	Acknowledges link state update
Database description	Announces which updates the sender has
Link state request	Requests information from the partner

The five types of OSPF messages.

BGP – The Exterior Gateway Routing Protocol



(a)

Information F receives
from its neighbors about D

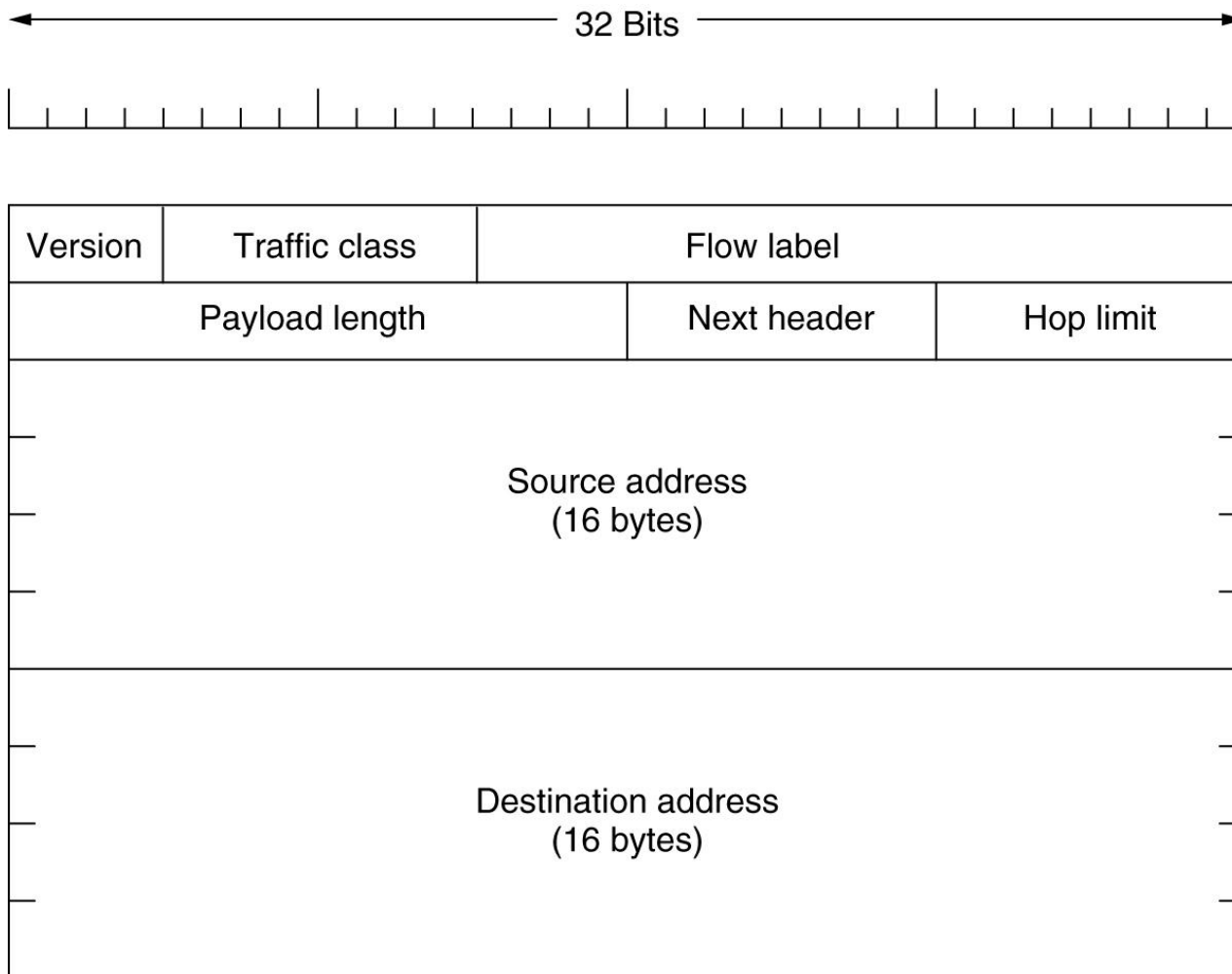
From B: "I use BCD"
From G: "I use GCD"
From I: "I use IFGCD"
From E: "I use EFGCD"

(b)

(a) A set of BGP routers.

(b) Information sent to F.

The Main IPv6 Header



The IPv6 fixed header (required).

Extension Headers

Extension header	Description
Hop-by-hop options	Miscellaneous information for routers
Destination options	Additional information for the destination
Routing	Loose list of routers to visit
Fragmentation	Management of datagram fragments
Authentication	Verification of the sender's identity
Encrypted security payload	Information about the encrypted contents

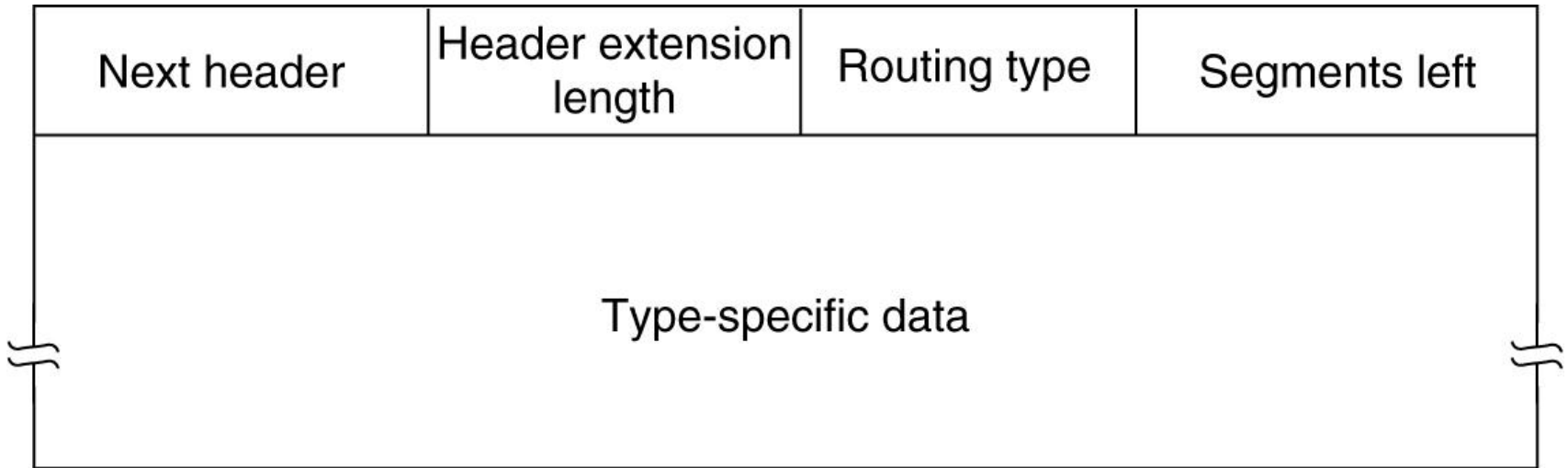
IPv6 extension headers.

Extension Headers (2)

Next header	0	194	4
Jumbo payload length			

The hop-by-hop extension header for large datagrams (jumbograms).

Extension Headers (3)



The extension header for routing.